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[Intervention Review]

Progesterone receptor modulators for endometriosis

Alexandros Loukas Grammatidis¹, Tatjana Gibbons², Eberechi N Anucha³, Christian M Becker⁴

¹Birmingham Women's Fertility Centre, Birmingham Women's Hospital NHS Foundation Trust, Birmingham, UK. ²Department of Gynaecology, Fertility and Obstetrics, Rigshospitalet, Copenhagen, Denmark. ³Oxford University Hospitals NHS Foundation Trust, Oxford, UK. ⁴Nuffield Department of Women's & Reproductive Health, University of Oxford, Oxford, UK

Contact: Alexandros Loukas Grammatidis, alexgrammatidis@gmail.com.

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ABSTRACT

Rationale

Endometriosis is defined as the presence of endometrium-like tissue developing outside the uterine cavity. This condition is oestrogen-dependent and thus is seen primarily during the reproductive years. Owing to their antiproliferative effects on the endometrium, progesterone receptor modulators (PRMs) have been advocated for the treatment of endometriosis. This is an update of a Cochrane review previously published in 2017.

Objectives

To assess the effectiveness and safety of progesterone receptor modulators (PRMs) in the management of endometriosis.

Search methods

To identify studies, we searched online sources, including the Cochrane Gynaecology and Fertility (CGF) Group trials register, CENTRAL, MEDLINE, Embase, PsycINFO, ClinicalTrials.gov and the WHO (World Health Organization) trial registry up to 16 April 2026. We also checked reference lists and contacted study authors and experts.

Eligibility criteria

We included randomised controlled trials (RCTs) comparing PRMs with placebo, no treatment or other medical treatment, in women with symptomatic endometriosis.

Outcomes

Our critical outcomes were overall pain; improvement in the most troublesome symptom, including dysmenorrhoea and dyspareunia; and adverse events resulting from the PRM intervention. Our important outcomes included quality of life (QoL) and patient satisfaction with treatment.

Risk of bias

We used the Cochrane tool RoB 2 to assess the risk of bias in the outcomes of this review.

Synthesis methods

Two review authors independently assessed studies against the inclusion criteria, extracted data and assessed risk of bias, consulting a third review author where required. We analysed dichotomous outcomes using Mantel-Haenszel odds ratios (ORs), 95% confidence intervals (CIs) and the fixed-effect model. We analysed continuous outcomes using the mean difference (MD) between groups, presented with 95% CIs. We employed the GRADE approach to assess the certainty of the evidence as high, moderate, low or very low.

Included studies

No new studies were eligible for inclusion in this update.

Of the nine studies (922 women) already included in the review, two compared mifepristone versus placebo or compared different doses of mifepristone, and four either compared gestrinone versus danazol, gestrinone versus gonadotropin-releasing hormone (GnRH) analogues or compared different doses of gestrinone. One study compared asoprisnil versus placebo. The asoprisnil study did not provide extractable numerical data and was included in the narrative synthesis only. Two of the studies did not report any of the review's prespecified outcomes.

Synthesis of results

Certainty of the evidence

We judged the certainty of evidence as low to very low. The main limitations of the evidence were serious risk of bias (associated with poor reporting of methods and high or unclear rates of attrition), very serious imprecision (associated with low event rates and wide confidence intervals, with some large effect estimates) and serious indirectness (outcomes assessed in selected subgroups).

PRM versus placebo

Mifepristone versus placebo

These results are based on one study (360 participants) that measured outcomes after three months.

The study did not report overall pain. At three months, mifepristone may improve dysmenorrhoea (OR 0.07, 95% CI 0.03 to 0.15; 1 study, 342 participants; low-certainty evidence), suggesting that if 40% of women taking placebo experience dysmenorrhoea, then between 3% and 10% of women taking mifepristone will do so. We are uncertain whether mifepristone improves dyspareunia (OR 0.23, 95% CI 0.10 to 0.51; 1 study, 223 participants; very low-certainty evidence). However, mifepristone may increase hot flushes (OR 28.79, 95% CI 3.93 to 210.73; 1 study, 360 participants; low-certainty evidence). Mifepristone may make little or no difference after three months of treatment to the prevalence of nausea (OR 1.72, 95% CI 0.20 to 15.03; 1 study, 360 participants; low-certainty evidence).

PRM versus other medical treatment

Gestrinone versus danazol

These results are based on two studies (302 participants) that measured outcomes after six months.

We are uncertain of the effects of gestrinone relative to danazol on the improvement of overall pain (dichotomous outcome: participants reporting no or mild pelvic pain counted as 'improved') (OR 0.47, 95% CI 0.04 to 5.70; 1 study, 38 participants), dysmenorrhoea (Not estimable; 1 study, 38 participants), or dyspareunia (OR 1.00, 95% CI 0.13 to 7.94; 1 study, 38 participants (all very low-certainty evidence)). We are uncertain about the effects of gestrinone relative to danazol on hot flushes (OR 0.79, 95% CI 0.50 to 1.26; 2 studies, 302 participants; very low-certainty evidence). There may be little or no difference between gestrinone and danazol in rates of nausea (OR 1.36, 95% CI 0.84 to 2.19; 2 studies, 302 participants; low-certainty evidence).

Gestrinone versus a GnRH analogue (leuporelin)

These results are based on one study (55 participants) that measured outcomes after six months.

There may be little to no difference between gestrinone and leuporelin in non-menstrual pain (MD -0.41, 95% CI -1.76 to 0.94; 1 study, 55 participants; low-certainty evidence). Women taking gestrinone may have higher dysmenorrhoea scores than those taking leuporelin (MD 0.82, 95% CI 0.15 to 1.49; 1 study, 55 participants; low-certainty evidence), but gestrinone may improve dyspareunia more than leuporelin (MD -1.16, 95% CI -2.08 to -0.24; 1 study, 52 participants; low-certainty evidence). There may be lower rates of hot flushes with gestrinone than leuporelin (OR 0.20, 95% CI 0.06 to 0.63; 1 study, 55 participants; low-certainty evidence). There may be little or no difference between gestrinone and leuporelin in rates of nausea (OR 1.04, 95% CI 0.06 to 17.49; 1 study, 55 participants; low-certainty evidence).

Authors' conclusions

Mifepristone may improve dysmenorrhoea in women with endometriosis, but it might increase the prevalence of adverse events, such as hot flushes. The evidence is very uncertain regarding its effect on dyspareunia.

Gestrinone may be less effective than leuporelin in treating dysmenorrhoea, but it may improve dyspareunia. It may be associated with lower rates of hot flushes and similar rates of nausea. We are uncertain about the effect of gestrinone compared to danazol.

We found insufficient evidence to permit firm conclusions about the safety and effectiveness of other progesterone receptor modulators.

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Registration

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Review (2017) DOI: 10.1002/14651858.CD009881.pub2

PLAIN LANGUAGE SUMMARY

What are the benefits and risks of medicines called progesterone receptor modulators (PRMs) for treating endometriosis (a condition where tissue grows outside the womb)?

Key messages

- In the management of endometriosis (a condition where tissue grows outside the womb), studies that compared a progesterone receptor modulator (PRM; a medicine that changes how the hormone progesterone acts in the body) against placebo (a 'dummy' treatment, or sham treatment, that does not contain any medicine but looks or tastes identical to the medicine being tested) found that a PRM called mifepristone may improve painful periods. However, it may cause hot flushes.
- A PRM called gestrinone may be less effective than medicines known as gonadotropin-releasing hormone (GnRH) analogues (which reduce the amount of oestrogen made by the ovaries) for treating painful periods, but may improve pain during sex. Gestrinone may also cause fewer hot flushes. We do not know how gestrinone compares to a treatment called danazol in terms of pain not related to periods, painful periods or pain during sex. We do not know whether gestrinone or danazol causes more unwanted effects.
- Future studies should compare progesterone receptor modulators with medicines currently used to treat endometriosis symptoms and focus on assessing pain using recognised and reliable methods. Larger, well-designed studies are also needed to assess the benefits and unwanted effects of progesterone receptor modulators other than mifepristone and gestrinone.

What is endometriosis?

Endometriosis is a condition where tissue similar to the lining of the womb (uterus) grows outside the womb. Endometriosis is oestrogen-dependent and thus is seen mainly during the reproductive years. It can cause pain in the tummy area (abdomen), generally during periods (menstruation) or associated with sexual intercourse.

Endometriosis is treated through different types of medical or surgical treatment. Progesterone receptor modulators (PRMs) are medicines that change how the hormone progesterone acts in the body. PRMs include mifepristone, gestrinone, ulipristal acetate, vilaprisan and asoprisnil. Because they can slow down or stop the growth of tissue in the lining of the uterus, they have been suggested as a treatment for endometriosis.

What did we want to find out?

We wanted to know whether, in patients with endometriosis, PRMs are better than placebo ('dummy' treatment), no treatment or other medical treatment for improving overall pain, pain during periods and pain during sexual intercourse.

We also wanted to know if these medications were safe by assessing unwanted effects experienced by patients, such as hot flushes and feeling sick (nausea).

What did we do?

We searched for studies that investigated the effectiveness of PRMs for symptoms associated with endometriosis and whether they caused unwanted effects. We compared and summarised the results of the studies and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We included nine studies involving 922 women aged 18 to 45 years. Most studies tested the PRMs mifepristone or gestrinone. One study tested asoprisnil, but the study was published only as a short summary without results that we could use. Two of the studies did not measure any of the effects we were interested in.

Main results

Progesterone receptor modulators versus placebo

When compared to a placebo, mifepristone:

- may reduce painful periods after three months of treatment (based on 1 study of 342 women);
- may cause more unwanted effects, such as hot flushes (1 study, 360 women);
- may make little or no difference to nausea (1 study, 360 women); and
- we are uncertain whether mifepristone reduces pain during sex (1 study, 223 women).

Progesterone receptor modulators versus other medical treatment

When compared to danazol, one study of 38 women showed that:

- we do not know whether gestrinone improves pain symptoms after six months of treatment (1 study, 38 women);
- we do not know whether gestrinone causes more or fewer hot flushes or makes no difference (2 studies, 302 women); and
- gestrinone and danazol may cause similar rates of nausea, but the results are very uncertain (2 studies, 302 women).

When compared to a GnRH analogue (leuporelin), one study of 55 women found that:

- gestrinone may be less efficient in treating painful periods after six months of treatment (1 study, 55 women); but
- gestrinone may reduce pain during sex (1 study, 52 women);
- gestrinone may be similar to leuporelin in terms of general pain experienced (1 study, 55 women);
- gestrinone may cause fewer hot flushes and similar levels of nausea (1 study, 55 women).

What are the limitations of the evidence?

Our confidence in the evidence is low to very low, mainly because some studies had one or more of these problems:

- they did not have many people taking part;
- they focused on specific settings or populations;
- they used methods likely to introduce errors in their results;
- they did not clearly report how they were conducted;
- they did not clearly report whether the people taking part knew which treatment they had received, which could have affected the study results.

How up to date is this evidence?

This review updates a previous review. The evidence is up to date to April 2026.

SUMMARY OF FINDINGS

Summary of findings 1. Mifepristone versus placebo for endometriosis

Mifepristone versus placebo for endometriosis

Patient or population: women with symptomatic endometriosis

Settings: gynaecology clinic

Intervention: progesterone receptor modulator (mifepristone)

Comparison: placebo

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Placebo	Mifepristone				
Overall pain	Not reported					
Dysmenorrhoea Follow-up: 3 months	402 per 1000	51 per 1000 (26 to 103)	OR 0.07 (0.03 to 0.15)	342 (1)	⊕⊕⊕⊖ Low ^{a,b}	Mifepristone may improve dysmenorrhoea after 3 months of treatment.
Dyspareunia Follow-up: 3 months	288 per 1000	85 per 1000 (43 to 171)	OR 0.23 (0.10 to 0.51)	223 (1)	⊕⊖⊖⊖ Very low ^{a,b,c}	We are uncertain whether mifepristone improves dyspareunia after 3 months of treatment.
Adverse events: hot flushes Follow-up: 3 months	11 per 1000	243 per 1000 (42 to 701)	OR 28.79 (3.93 to 210.73)	360 (1)	⊕⊕⊖⊖ Low ^d	Mifepristone may have higher rates of hot flushes after 3 months of treatment.
Adverse events: nausea Follow-up: 3 months	0 per 1000	0 per 1000 (0 to 0)	OR 1.72 (0.20 to 15.03)	360 (1)	⊕⊕⊖⊖ Low ^d	Mifepristone may make little or no difference to nausea after 3 months of treatment.

*The basis for the **assumed risk** is the mean control group risk across studies. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI)
CI: confidence interval; OR: odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aData derived from a single study. Downgraded one level for serious imprecision

^bThe authors use two methods of evaluating pain (severity indicated by visual analogue scale and symptom prevalence) and appear to focus on the favourable result. Downgraded for selective reporting

^cOutcome applied only to women with dyspareunia at baseline, which was 223/360 women randomised. Downgraded one level for serious indirectness

^dData derived from a single study. Downgraded two levels for serious imprecision (wide confidence intervals and/or very few events)

Summary of findings 2. Gestrinone versus danazol for endometriosis

Gestrinone versus danazol for endometriosis

Patient or population: women with symptomatic endometriosis

Settings: gynaecology clinic

Intervention: progesterone receptor modulator (gestrinone)

Comparison: danazol

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Danazol	Gestrinone				
Pelvic pain: none or mild Follow-up: 6 months	947 per 1000	894 per 1000 (419 to 990)	OR 0.47 (0.04 to 5.70)	38 (1)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence concerning the effect of gestrinone compared to danazol on overall pain after 6 months of treatment is very uncertain.
Dysmenorrhoea: none or mild Follow-up: 6 months	Not estimable	Not estimable	Not estimable	38 (1)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence concerning the effect of gestrinone compared to danazol on dysmenorrhea after 6 months of treatment is very uncertain.

Dyspareunia: none or mild	895 per 1000	895 per 1000 (525 to 985)	OR 1.00 (0.13 to 7.94)	38 (1)	⊕⊕⊕⊕ Very low ^{a,b}	The evidence concerning the effect of gestrinone compared to danazol on dyspareunia after 6 months of treatment is very uncertain.
Follow-up: 6 months						
Adverse events: hot flushes	425 per 1000	368 per 1000 (270 to 482)	OR 0.79 (0.50 to 1.26)	302 (2)	⊕⊕⊕⊕ Very low ^{b,c,d}	The evidence concerning the effect of gestrinone compared to danazol on hot flushes after 6 months of treatment is very uncertain.
Follow-up: 6 months						
Adverse events: nausea	503 per 1000	579 per 1000 (460 to 689)	OR 1.36 (0.84 to 2.19)	302 (2 studies)	⊕⊕⊕⊕ Low ^{c,d}	There may be little to no difference in rates of nausea between gestrinone groups and danazol groups after 6 months of treatment.
Follow-up: 6 months						

*The basis for the **assumed risk** is the mean control group risk across studies. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI)
CI: confidence interval; OR: odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

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Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aData derived from a single study. Small number of events and wide confidence intervals. Downgraded twice for very serious imprecision

^bDowngraded one level for serious risk of bias associated with measurement of the outcome

^cDowngraded one level for serious risk of bias associated with poor reporting of study methods, high attrition in one study and high risk of other bias in both studies

^dDowngraded one level for serious inconsistency

Summary of findings 3. Gestrinone versus a GnRH analogue (leuprorelin) for endometriosis

Gestrinone versus a GnRH analogue (leuprorelin) for endometriosis

Patient or population: women with symptomatic endometriosis

Settings: gynaecology clinic

Intervention: progesterone receptor modulator (gestrinone)

Comparison: GnRH analogue (leuprorelin)

Outcomes	Illustrative comparative risks* (95% CI)		Relative effect (95% CI)	Number of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Assumed risk	Corresponding risk				
	Leuprorelin	Gestrinone				
Non-menstrual pain, visual analogue scale Follow-up: 6 months	The mean non-menstrual pain, visual analogue scale was 1.64 .	MD 0.41 lower (1.76 lower to 0.94 higher)	-	55 (1)	⊕⊕⊕⊕ Low ^a	There may be little to no difference in non-menstrual pain between gestrinone and leuprorelin groups after 6 months of treatment.
Dysmenorrhoea, visual analogue scale Follow-up: 6 months	The mean dysmenorrhoea, visual analogue scale was 0.05 .	MD 0.82 higher (0.15 higher to 1.49 higher)	-	55 (1)	⊕⊕⊕⊕ Low ^a	Gestrinone may be associated with worse rates of dysmenorrhoea than leuprorelin after 6 months of treatment.
Dyspareunia, visual analogue scale Follow-up: 6 months	The mean dyspareunia, visual analogue scale was 1.6 .	MD 1.16 points lower (2.08 to 0.24 lower)	-	52 (1)	⊕⊕⊕⊕ Low ^a	Gestrinone may improve dyspareunia more than leuprorelin after 6 months of treatment.
Adverse events: hot flushes Follow-up: 6 months	679 per 1000	297 per 1000 (112 to 571)	OR 0.20 (0.06 to 0.63)	55 (1)	⊕⊕⊕⊕ Low ^b	Gestrinone may be associated with lower rates of hot flushes than leuprorelin after 6 months of treatment.
Adverse events: nausea Follow-up: 6 months	36 per 1000	37 per 1000 (2 to 393)	OR 1.04 (0.06 to 17.49)	55 (1)	⊕⊕⊕⊕ Low ^b	There may be little to no difference in rates of nausea between gestrinone and leuprorelin groups after 6 months of treatment.

*The basis for the **assumed risk** is the mean control group risk across studies. The **corresponding risk** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).
CI: confidence interval; OR: odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

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Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aData available from one study with a small sample size. Downgraded two levels for very serious imprecision: confidence intervals were compatible with no clinically meaningful difference between groups, or with small benefit in one group.

^bDowngraded two levels for very serious imprecision: small overall sample size (n = 55) and low event rates

BACKGROUND

Description of the condition

Endometriosis is the presence of endometrial tissue (glands and stroma) outside the uterine cavity. It is a condition that is oestrogen-dependent and thus is seen primarily during the reproductive period. However, there are instances where symptoms persist beyond menopause [1, 2, 3]. The prevalence of endometriosis ranges from 6% to 10% in women of reproductive age, from 50% to 60% in women and teenage girls with pelvic pain, and has been reported as up to 50% amongst women with infertility [2, 4, 5]. Endometriosis can also be found in asymptomatic women during surgical procedures such as laparoscopy or sterilisation. Endometriosis constitutes a substantial burden on the health-related quality of life of women and on the finite healthcare resources of national health systems worldwide [6].

The condition was first described by the Austrian pathologist Carl von Rokitansky in 1860 [7]. Although considerable progress has been made in understanding the condition, the pathogenesis remains unclear. The 'retrograde menstruation theory' proposed in the 1920s speculated that endometriosis derives from reflux of endometrial fragments regurgitated through the fallopian tubes during menstruation, with subsequent implantation on the peritoneum and the ovary [8]. Other potential mechanisms of action include coelomic metaplasia, Mullerian remnant abnormalities, angiogenetic/lymphogenetic spread, metaplasia theory, and genetic/epigenetic theory [6, 9]. Hormonal imbalance involved in the development of endometriosis includes increased oestrogen synthesis and metabolism, and progesterone resistance [10]. Lastly, an increasing body of evidence supports the significant role of the immune system in the development and progression of endometriosis [11].

Diagnosis is usually based on symptoms together with clinical examination and imaging, although laparoscopy with histological confirmation has traditionally been regarded as the reference standard. Minimally invasive surgical treatments play an important role in the diagnosis and removal of endometriosis. A Cochrane review published in 2020 indicated that laparoscopic surgery may improve viable intrauterine pregnancy rates confirmed by ultrasound in women with endometriosis [12]. However, its effectiveness in reducing overall pain across the spectrum of disease severity—from minimal to severe—remains uncertain. Moreover, the recurrence of pain following surgical intervention is common, with over 60% of individuals diagnosed with endometriosis undergoing more than one surgical procedure [13].

The goals of medical management are to provide pain relief, to limit recurrence and to restore or preserve fertility when needed. Current medical treatment has focused on blocking ovarian oestrogen secretion (with gonadotropin-releasing hormone analogues or antagonists (GnRH-a or GnRH antagonists)) or halting oestrogen-induced growth of ectopic endometrium (with oral contraceptives, androgens and aromatase inhibitors) [14]. Previous Cochrane Reviews have reported that progestins [15], GnRH agonists [16] and danazol [17] appear to be effective for reducing pain symptoms associated with endometriosis. However, the clinical use of danazol has significantly declined in recent years due to its unfavourable side effect profile [17]. Evidence regarding the efficacy of combined oral contraceptives remains inconclusive [18]. At present, there is insufficient high-quality evidence to support the routine use

of aromatase inhibitors for pain management in endometriosis. Nonetheless, the European Society of Human Reproduction and Embryology (ESHRE) 2022 guidelines recommend that, in cases of pain due to drug- and surgery-resistant recto-vaginal endometriosis, aromatase inhibitors may be used in combination with oral contraceptives, progestogens or GnRH agonists [3].

The side effect profiles of conventional hormone-based therapies for endometriosis present a significant challenge in balancing therapeutic efficacy with patient safety [17, 19]. Combined oral contraceptives are commonly associated with adverse effects such as nausea, headache, breast tenderness, weight gain and breakthrough bleeding. Progestogens may lead to irregular menstrual bleeding, weight gain, mood disturbances and reduced libido [15]. Gonadotropin-releasing hormone (GnRH) agonists induce a marked suppression of circulating oestrogen levels, often resulting in hypo-oestrogenic side effects, including vasomotor symptoms and loss of bone mineral density with prolonged use—though these effects are typically reversible upon cessation of treatment [16]. Aromatase inhibitors have similarly been linked to an increased risk of osteoporosis and bone fractures [20]. Danazol, though infrequently used today as mentioned above, is associated with androgenic side effects such as acne, weight gain and, in some cases, irreversible deepening of the voice [17].

The mean length of time before recurrence of pain after completion of medical therapy is between 6 and 18 months [21]. The ideal medical therapy for endometriosis should relieve pain and inhibit endometrial proliferation while avoiding a hypo-oestrogenic state.

Description of the intervention and how it might work

Progesterone, acting through progesterone receptors, plays a central role in regulating uterine and endometrial function, making the progesterone receptor a potential therapeutic target for endometriosis. Progesterone receptor modulators (PRMs) are synthetic ligands that bind to progesterone receptors and may exert agonist, antagonist or mixed agonist–antagonist effects. Their activity depends on the pharmacological properties of the compound, the relative expression of progesterone receptor subtypes, and the balance of receptor co-activators and co-repressors within individual tissues. Progesterone receptor antagonists (PRAs) predominantly inhibit progesterone activity, whereas selective progesterone receptor modulators (SPRMs) exert differing combinations of agonist and antagonist effects in different tissues. Thus, PRAs and SPRMs form part of a broader spectrum of progesterone receptor ligands rather than representing entirely separate drug classes [22].

Mifepristone (MFP; RU486), first synthesised in 1980, acts predominantly as an antagonist at progesterone and glucocorticoid receptors. More than 100 trial-like studies from China have examined its use for endometriosis. However, substantial limitations in their design, conduct and reporting mean that they provide no reliable conclusions regarding its indications, dosage, efficacy or safety [23]. Other progesterone receptor ligands have subsequently been developed with varying degrees of agonist and antagonist activity. Onapristone, lonaprisan, vilaprisan and mifepristone exhibit predominantly antagonistic effects, whereas asoprisnil has a comparatively greater agonist component. Other agents, including ulipristal acetate and telapristone acetate, lie at intermediate points along this agonist–antagonist spectrum [22, 24].

Gestrinone is a synthetic 19-norsteroid with antiprogesterogenic, anti-oestrogenic and androgenic activity. Its combined hormonal effects suppress ovarian steroidogenesis and endometrial proliferation, which led to its use in the treatment of oestrogen-dependent conditions such as endometriosis [25].

PRMs have been proposed as treatments for symptomatic endometriosis because of their antiproliferative effects on the endometrium and endometriotic tissue. However, they may not be appropriate for treating asymptomatic endometriosis-associated infertility because they can inhibit ovulation and produce amenorrhoea [26, 27]. Endometriosis is characterised by an aberrant response to progesterone that contributes to an oestrogen-rich, proinflammatory environment [28]. By antagonising progesterone-mediated signalling, PRMs may inhibit endometrial proliferation while maintaining circulating oestradiol concentrations. Therefore, unlike gonadotropin-releasing hormone agonists, they may avoid the systemic effects of profound oestrogen deprivation, including loss of bone mineral density [29].

Mifepristone is licensed in many countries, in combination with a prostaglandin analogue, for medical termination of pregnancy. The single higher dose used for this indication differs substantially from the lower daily doses and longer treatment durations evaluated for endometriosis [30, 31]. Ulipristal acetate is licensed as a single-dose emergency contraceptive. Neither asoprisnil nor the other investigational SPRMs discussed in this review have been approved as an emergency contraceptive. PRMs have also been shown to be effective in the management of uterine fibroids [32]. However, the use of ulipristal acetate 5 mg for uterine fibroids has been substantially restricted because of reports of rare but serious liver injury [33].

Few studies have evaluated PRMs for endometriosis, although some have reported potential clinical benefits [34, 35]. In an uncontrolled study, mifepristone relieved pelvic pain without causing significant adverse effects [36]. Mifepristone has also been reported to downregulate progesterone and oestrogen receptors and upregulate androgen receptors [37]. In human endometrial cell lines, mifepristone, asoprisnil, ulipristal acetate, lonaprisan and telapristone acetate suppressed cellular proliferation and promoted endometrial atrophy [29]. In animal models, mifepristone, onapristone and ZK136799 suppressed endometrial growth and reduced prostaglandin production, which may contribute to pain relief [38]. PRMs may therefore reduce endometriosis-related pain by selectively inhibiting endometrial proliferation, decreasing menstruation, and suppressing endometrial prostaglandin production without producing the marked systemic hypo-oestrogenism associated with some conventional treatments [10].

PRM administration may produce distinctive morphological changes in the endometrium, known as PRM-associated endometrial changes (PAECs) [39]. The cause of the associated cystic changes remains uncertain, although proposed mechanisms include apoptosis [40] and collapse of stromal vessels [41], potentially leading to tissue necrosis and obstruction of uterine gland openings. Consensus studies describe PAECs as benign and reversible [39, 42]. Nevertheless, the clinical effectiveness and safety of PRMs for endometriosis require further evaluation.

Why it is important to do this review

Doctors considering the use of PRMs in clinical practice are uncertain about the balance between benefit in terms of pain relief and unacceptable side effects, particularly in the context of growing evidence for progesterone resistance in endometriosis, which is thought to be mediated through multifactorial downregulation of progesterone receptors [22, 28, 43]. In this review, we evaluated the clinical effectiveness and safety of PRMs, primarily in terms of pain relief, and collated the best evidence for their use as single agents in women with endometriosis.

OBJECTIVES

To assess the effectiveness and safety of progesterone receptor modulators (PRMs) in the management of endometriosis.

METHODS

We have made a range of changes to the protocol for this review update, in terms of the title, objectives, inclusion criteria, and data analysis methods. We have documented these differences in [Supplementary material 9](#).

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs), published in any language, that examined the effects of progesterone receptor modulators (PRMs) for the treatment of symptomatic endometriosis. We excluded quasi-RCTs. We also excluded cross-over trials, as the design is not valid in this context.

Types of participants

The participants were women with symptomatic endometriosis. For a trial to be included in the review, the clinical diagnosis of endometriosis must have been made by direct visualisation (laparoscopy or laparotomy) or from ultrasonographic imaging or magnetic resonance imaging (MRI). We did not consider trials that included women with asymptomatic disease or infertility alone. For studies that included both symptomatic and asymptomatic women, we planned to include data only where outcomes for women with symptomatic endometriosis were reported separately or could be obtained from the study authors.

Types of interventions

Intervention

PRM for any period of time

Comparison

- Placebo
- No treatment
- Other medical therapies (combined oral contraceptive pill, GnRH analogues, progestogens, antiprogestogens, levonorgestrel-releasing intrauterine systems, aromatase inhibitors)
- Another dose or regimen of the intervention PRM

We did not consider studies involving surgical removal of pelvic organs, or studies exploring the use of hormonal treatment as an

adjunct to therapeutic surgery or to another medical treatment for endometriosis.

Outcome measures

A minimum data set, known as a core outcome set, has been developed for endometriosis research using formal consensus methods involving 116 healthcare professionals, 32 researchers, and 206 women with endometriosis from 29 countries [44]. For this update, we selected the review outcomes based on that consensus.

We analysed outcomes at the end of treatment or at the nearest reported time point where multiple time points were available; where data were reported at several time points, we prioritised clinically relevant follow-up durations.

Critical outcomes

- Overall pain measured on a visual analogue scale (VAS) of pain, other validated scales, or dichotomous outcomes.
- Improvement in the most troublesome symptom, including dysmenorrhoea and dyspareunia, measured using validated pain scales (e.g. visual analogue scale (VAS), verbal rating scales, or Biberoglu & Behrman score [45] or as dichotomous outcomes (e.g. improvement versus no improvement), as defined by the study authors.
- Adverse events resulting from the PRM intervention (including hot flushes, irregular bleeding, headache, nausea, anorexia, vomiting, fatigue, allergic reaction, acne, hirsutism, amenorrhoea, liver toxicity, endometrial hyperplasia, PRM-associated endometrial changes (PAEC), or hypoadrenalism).

Important outcomes

- Quality of life (QOL). If studies report the use of more than one scale, we gave preference to the Short Form (SF)-36, then to other validated generic scales, and finally to condition-specific scales.
- Patient satisfaction with treatment, measured using validated scales or questionnaires (e.g. Likert scales or patient-reported outcome measures), or as defined by the study authors.

Search methods for identification of studies

Electronic searches

We developed a comprehensive literature search strategy in consultation with the Information Specialist of the Cochrane Gynaecology and Fertility Group (CGFG). Two review authors (ALG and TG) independently conducted a systematic search of the published and unpublished literature from inception of the databases to 8 August 2024 with no restrictions on language or publication status as per Cochrane criteria [46]. We conducted an updated search on 16 April 2026 using the same search strategy and databases.

We searched the following databases.

- CGFG Specialised Register (from inception to 16 April 2026) ([Supplementary material 1](#))
- Cochrane Central Register of Controlled Trials (CENTRAL; 2026, Issue 4) via CRS Web ([Supplementary material 1](#))

- Ovid MEDLINE(R) In-Process & Other Non-Indexed Citations, Ovid MEDLINE(R) Daily, and Ovid MEDLINE(R) (Ovid; from 1946 to 16 April 2026) ([Supplementary material 1](#))
- Ovid Embase (Ovid; from 1974 to 16 April 2026) ([Supplementary material 1](#))

The search strategies incorporated validated methodological filters for identifying randomised controlled trials, as detailed in [Supplementary material 1](#).

Searching other resources

We scanned the references of all included studies and relevant reviews to identify additional relevant trials, and we handsearched relevant journals and articles that may not be included in the electronic databases. We handsearched relevant journals and conference abstracts not covered by the Cochrane Gynaecology and Fertility Specialised Register, in liaison with the Group's Information Specialist, Marian Showell. We also scanned the following databases from inception to August 2024, and updated the searches on 16 April 2026.

- ClinicalTrials.gov (<https://clinicaltrials.gov/>)
- World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (<https://trialsearch.who.int/>)
- DARE (Database of Abstracts of Reviews of Effects) in the Cochrane Library for reference lists from relevant non-Cochrane reviews
- OpenGrey (<http://www.opengrey.eu/>) for unpublished European literature
- LILACS (Latin American and Caribbean Health Sciences Literature)
- PubMed and Google Scholar for recently published studies not yet indexed in MEDLINE

We also checked included and potentially eligible studies for post-publication amendments, including retractions, expressions of concern, errata, and corrigenda, in accordance with current Cochrane guidance.

Data collection and analysis

Selection of studies

All titles and abstracts retrieved by electronic searching were downloaded to a reference management database and duplicates were automatically removed; two review authors (ALG and TG) independently examined the remaining references. We excluded studies that clearly did not meet the inclusion criteria and obtained copies of the full text of potentially relevant references. Two review authors (ALG and TG) independently assessed the eligibility of retrieved papers. Review authors resolved differences by discussion or by appeal to a third review author (CB). We documented reasons for exclusion and presented the selection process in a PRISMA flow chart [47].

Data extraction and management

Two review authors (ALG and TG) independently extracted data from eligible studies using a data extraction form that they designed and pilot-tested.

Extracted data included:

- participant characteristics;
- number of participants in each arm of the trial;
- number excluded from analysis;
- type of intervention;
- proportion of participants who received all/part/none of the intended treatment;
- methods of randomisation, blinding, and allocation concealment applied;
- length of follow-up; and
- outcome measurements.

We resolved differences between review authors by discussion or by appeal to a third review author (CB), if necessary. Review authors were not blinded to article author or journal title. We tried to contact trial authors for further information and updated data where necessary.

Risk of bias assessment in included studies

Two review authors (ENA or ALG) independently assessed the methodological risk of bias of each trial using the revised risk of bias (RoB 2) tool, as per the Cochrane guidelines [48]. In the Cochrane RoB 2 tool, assessment for risk of bias in included studies consists of five domains: risk of bias arising from the randomisation process; risk of bias due to deviations from intended interventions; risk of bias due to missing outcome data; risk of bias in measurement of the outcome; risk of bias in selection of reported result.

For each domain, we applied the RoB 2 signalling questions and guidance. For bias arising from the randomisation process, we judged low risk where sequence generation was adequate (e.g. computer-generated randomisation, random number tables) and allocation concealment robust (e.g. central randomisation, sequentially numbered sealed opaque envelopes), and high risk where inadequate or open allocation methods were used. For bias due to deviations from intended interventions, we considered whether participants and study personnel were blinded, and whether placebos were used. For bias due to missing outcome data, we considered the proportion of participants with missing outcome data, whether missingness differed between intervention groups, the reasons for missing data, and whether missingness could have depended on the true value of the outcome. We applied the RoB 2 signalling questions and algorithm to reach the domain-level judgement rather than using a prespecified numerical threshold for attrition. For bias in measurement of the outcome, we assessed blinding of outcome assessors and use of validated measurement tools. For bias in selection of the reported result, we compared published reports with protocols or trial registry entries to identify selective reporting. Where information was insufficient to reach a judgement, we sought clarification from study authors.

We made domain-level judgements as “low risk of bias,” “some concerns,” or “high risk of bias.” We then made an overall risk of bias judgement for each outcome, following the RoB 2 algorithm:

- overall “low risk of bias” if all domains were judged low risk;
- overall “some concerns” if at least one domain raised some concerns but none were high risk;
- overall “high risk of bias” if one or more domains were judged high risk, or if multiple domains raised some concerns.

Our primary interest was in estimating the effect of assignment to the intervention (intention-to-treat effect), regardless of adherence. We assessed the risk of bias for the prespecified outcomes: pain scores (measured on validated scales such as VAS or NRS, at short-term and long-term time points), improvement in the most-troublesome symptom, adverse events (as defined and reported in the trials), health-related quality of life (validated questionnaires, short term and long term), and patient satisfaction with treatment.

We made no adaptations to the RoB 2 tool [48]. Any disagreements between review authors were resolved by discussion or, if necessary, by consultation with the Cochrane Gynaecology and Fertility Group (CGFG). Review authors with direct involvement in the conduct, analysis, or publication of any included trial did not participate in eligibility decisions, data extraction, risk of bias assessment, or GRADE judgements for that study, in accordance with Cochrane editorial policy on conflicts of interest.

Measures of treatment effect

For dichotomous data (e.g. dichotomous pain outcomes), we used numbers of events in control and intervention groups of each study to calculate Mantel-Haenszel odds ratios. For continuous outcomes measured on the same scale, we used the mean difference (MD) with 95% confidence intervals (CIs). If similar outcomes were reported on different scales, we calculated standardised mean differences (SMDs) and interpreted effect sizes as small (0.2), moderate (0.5) or large (0.8). We treated continuous or quasi-continuous outcomes (e.g. pain scores measured on visual analogue scales) as continuous data and presented 95% CIs for all outcomes [49].

Unit of analysis issues

We conducted the primary analysis per woman randomised.

For cluster-randomised trials, we planned to adjust for clustering using the methods described in Chapter 23 of the *Cochrane Handbook for Systematic Reviews of Interventions* [50], if required. For trials with multiple intervention arms (e.g. different doses of the same PRM), we included relevant intervention groups and, where appropriate, combined groups or split the shared comparator group to avoid double-counting participants, in accordance with Chapter 23 of the *Cochrane Handbook* [50]. Where appropriate, we analysed dose comparisons separately and described them narratively.

Dealing with missing data

We analysed data on an intention-to-treat basis as far as possible and attempted to obtain missing data from the original investigators [51]. If studies reported sufficient detail for calculation of mean differences but no information on associated standard deviation (SD), we planned to assume that outcomes had a standard deviation equal to the highest standard deviation used for other studies within the same analysis. Otherwise, we analysed only available data. We found that no imputation was necessary.

Reporting bias assessment

In view of the difficulty involved in detecting and correcting for publication bias and other reporting biases, we aimed to minimise their potential impact by ensuring a comprehensive search for eligible studies and by staying alert for duplication of data. If we

included 10 or more studies in an analysis, we planned to use a funnel plot to explore the possibility of a small-study effect (tendency for estimates of the intervention effect to be more beneficial in smaller studies).

Synthesis methods

We considered the following comparisons.

We combined data from primary studies evaluating the same outcome using the fixed-effect model for the following.

- PRMs versus placebo, stratified by dose
- PRMs versus no treatment, stratified by dose
- PRMs versus other medical therapies, stratified by dose (danazol, GnRH analogue, combined oral contraceptive pill (OCP), levonorgestrel-releasing intrauterine system, each in a separate analysis, not stratified)
- Dose or regimen comparison of the same PRM

Where studies reported improvement using different outcome measures, we used the definitions provided by the study authors and prioritised validated pain scales where available.

In the meta-analyses, we displayed graphically to the right of the centre line an increase in the odds of a particular outcome that may be beneficial (e.g. pain relief) or detrimental (e.g. adverse effects), and displayed to the left of the centre line a decrease in the odds of a particular outcome [52].

Investigation of heterogeneity and subgroup analysis

We assessed heterogeneity between studies by visual inspection of forest plots and by using the χ^2 test for heterogeneity (with a significance level of $P < 0.1$) and the I^2 statistic. We interpreted I^2 values in accordance with Chapter 10 of the *Cochrane Handbook* [52]: 0% to 40% might not be important; 30% to 60% may represent moderate heterogeneity; 50% to 90% may represent substantial heterogeneity; and 75% to 100% may represent considerable heterogeneity. Where substantial heterogeneity was identified, we explored possible reasons. We did not combine results of trials using different comparator drugs.

We planned to explore potential sources of heterogeneity by conducting subgroup analyses based on intervention characteristics (e.g. dose of PRM). Where studies included multiple dosing arms, we described possible dose-response patterns narratively and presented detailed dose data in the supplementary materials.

Equity-related assessment

We did not assess equity in this review.

Sensitivity analysis

We planned to conduct sensitivity analyses for the critical outcomes to determine whether conclusions were robust to

arbitrary decisions made regarding trial eligibility and analysis of results. Through these analyses, we planned to consider whether conclusions would have differed if:

- we had restricted eligibility to studies without high risk of bias;
- we had excluded studies with outlying results; or
- we had adopted the random-effects model.

Certainty of the evidence assessment

We prepared 'Summary of findings' tables using GRADEpro (Summary of findings 1; Summary of findings 2; Summary of findings 3). Two summary of findings tables of dose comparisons of mifepristone and gestrinone are presented in supplementary materials (Supplementary material 10; Supplementary material 11). The summary of findings tables present our evaluation of the overall certainty of the body of evidence for the critical review outcomes (i.e. overall pain (or pain as reported by authors), improvement in the most-troublesome symptom (dysmenorrhoea and dyspareunia), adverse events (hot flushes and nausea) for the main review comparisons (PRM versus placebo, PRM versus medical treatment (which we presented in two subcategories according to type of medical treatment comparator)). We assessed the certainty of the evidence using GRADE criteria: risk of bias, consistency of effect, imprecision, indirectness, and publication bias [53]. Two review authors independently judged the evidence certainty as high, moderate, low, or very low, and we resolved any disagreements by discussion. We justified our judgements and documented and incorporated them into the reporting of results for each outcome [54].

Consumer involvement

We did not involve consumers in this review. However, consumers were involved in the development of the core outcomes.

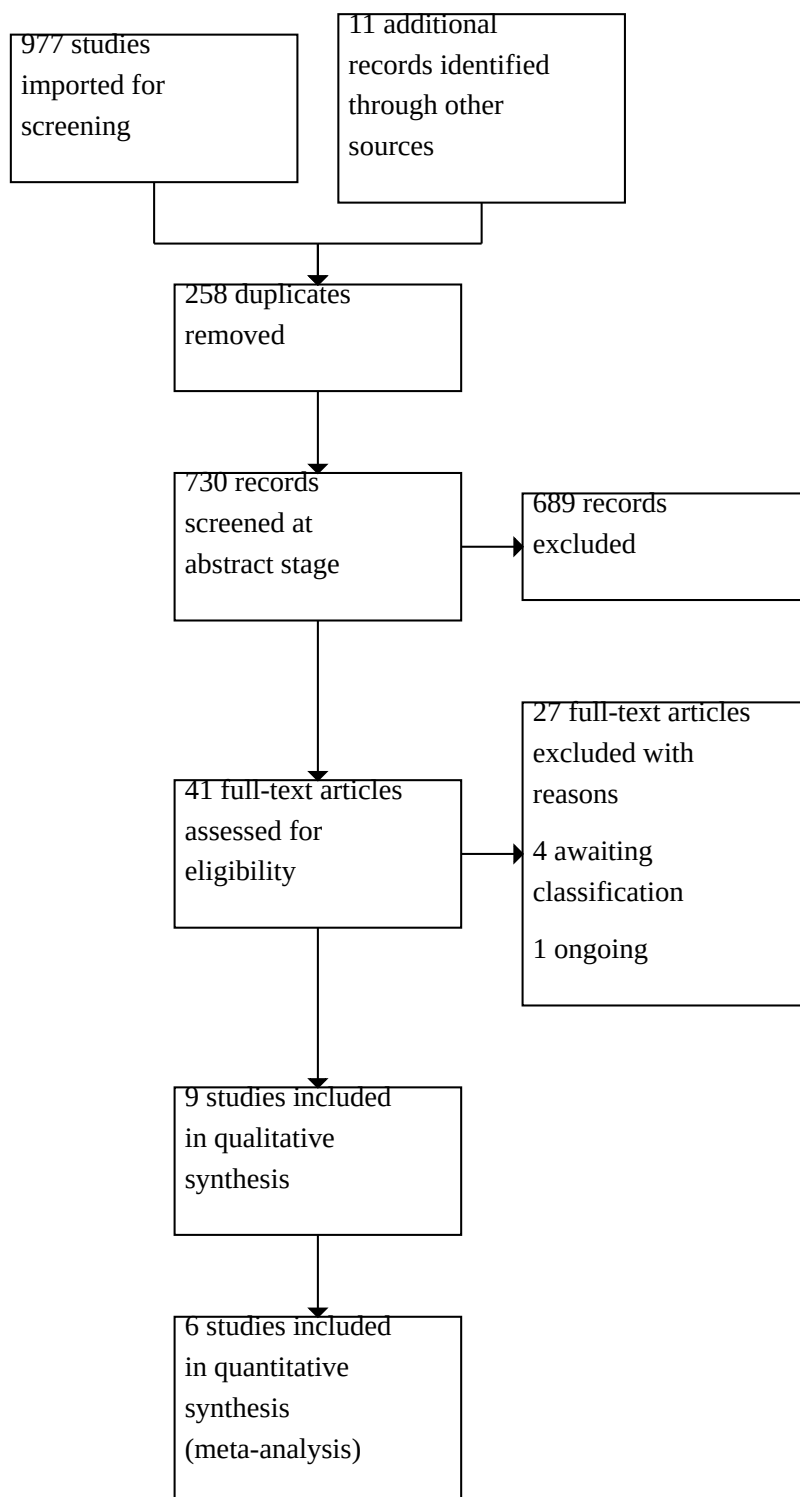
RESULTS

Description of studies

Results of the search

Following an updated search conducted on 16 April 2026, we screened 730 records and excluded 689 records at the title and abstract stage; we assessed 41 full-text records for eligibility. Nine studies met our inclusion criteria. We excluded 27 studies, four are awaiting classification, and one is ongoing. No new studies were identified for inclusion. No new studies were identified for inclusion. We excluded one study that had been included in the previous version of the review because we could not confirm that it was a randomised controlled trial (Spitz 2009 [55, 56]). The study was reported only as a book chapter, with no published abstract or full text available and no information on the method of randomisation. See Supplementary material 2, Supplementary material 3, Supplementary material 4, Supplementary material 5 and Figure 1.

Figure 1. Study flow diagram.



We attempted to contact trial authors to obtain additional information, including missing outcome data. We reported the data that were unsuitable for analysis in [Table 1](#). Of the nine included studies, seven reported one or more of the review's prespecified critical or important outcomes, and six contributed data to meta-analysis. We included one study, which evaluated asoprisnil versus placebo, in the narrative synthesis only (Chwalisz 2004 [34, 57]), because no extractable numerical data were available despite our attempts to contact the study authors. Two studies met the inclusion criteria in terms of study design, participants, and interventions, but did not report any prespecified review outcomes and therefore did not contribute to the synthesis (Dawood 1997 [58]; Worthington 1993 [59]).

Included studies

An overview of included studies is presented in [Table 2](#).

Study design and setting

We included nine parallel-design RCTs in the review, eight of which are published as full articles (Bromham 1995 [60, 61]; Carbonell 2012 [62]; Carbonell 2016 [63]; Fedele 1989 [64]; GISG 1996 [65]; Hornstein 1990 [66]; Dawood 1997; Worthington 1993) and one that is published only as an abstract (Chwalisz 2004). As mentioned above, two of the included studies did not report the critical and important outcomes of our review (Dawood 1997; Worthington 1993).

Two of the studies were multicentre studies: Bromham 1995 in the UK and GISG 1996 in Italy. Two single-centre studies were carried out in the USA (Dawood 1997; Hornstein 1990), two in Cuba (Carbonell 2012; Carbonell 2016), one in the UK (Worthington 1993) and one in Italy (Fedele 1989). In one study (Chwalisz 2004), the location was not specified. Two studies were conducted by one research group, differing in the year of study and sample size (Carbonell 2012; Carbonell 2016).

Participants

The nine included studies involved a total of 922 women: 545 in the intervention group and 308 in the control group. Sixty-nine women were involved in studies comparing different dosages of SPRMs, with no control group. All of the included participants had a diagnosis of endometriosis at laparoscopy, along with infertility in three studies (Fedele 1989; Hornstein 1990; Worthington 1993). The studies focused on women with varying American Fertility Society (AFS) scores. The age of participants was reported in all of the included studies and ranged from 18 to 45 years.

Interventions

Carbonell 2016 compared three different dosages of mifepristone with placebo. Carbonell 2012 compared different dosages of mifepristone for laparoscopically confirmed endometriosis. Chwalisz 2004 compared three doses of asoprisnil with placebo. Bromham 1995 and Fedele 1989 compared gestrinone with danazol. GISG 1996 compared gestrinone with leuprolerin (a gonadotropin-releasing hormone (GnRH) analogue). Hornstein 1990, Dawood 1997 and Worthington 1993 compared different dosages of gestrinone.

Outcomes

Seven studies reported one or more of our critical and important outcomes. Outcome data were reported only for participants who were symptomatic at baseline.

Critical outcomes

GISG 1996 reported on non-menstrual pain, dysmenorrhoea and dyspareunia using a visual analogue pain scale (VAS) and a verbal rating scale (Biberoglu & Behrman [45]). Carbonell 2012 reported dysmenorrhoea and dyspareunia using VAS, but the pain data were presented graphically without sufficient numerical data for meta-analysis; we therefore present them narratively in [Table 1](#) (see [Supplementary material 2](#) for further details). Chwalisz 2004 reported non-menstrual pain, dysmenorrhoea and dyspareunia using a verbal rating scale (Biberoglu & Behrman [45]), but the abstract did not provide extractable numerical outcome data; we therefore present the findings narratively in [Table 1](#) (see [Supplementary material 2](#)). Carbonell 2016, Fedele 1989 and Bromham 1995 reported on prevalence or severity of pelvic pain, dysmenorrhoea or dyspareunia compared to baseline, and Hornstein 1990 reported on improvement of pain. Bromham 1995 reported pain outcomes as percentages, but the denominators used to calculate these percentages were unclear because participant numbers varied across follow-up assessments; we therefore excluded these data from meta-analysis and instead describe them narratively in [Table 1](#) (see [Supplementary material 2](#) for further details).

Six studies reported on adverse events resulting from the SPRM intervention (Bromham 1995; Carbonell 2012; Carbonell 2016; Fedele 1989; GISG 1996; Hornstein 1990). Chwalisz 2004 did not report on adverse events.

Important outcomes

None of the included studies reported on patient satisfaction with treatment or quality of life.

Excluded studies

We excluded 27 studies from the review, including one additional study identified in the updated searches (Bisal 2025 [67]). Bisal 2025, Bulun 2016 [68], Bouchard 2011 [69], Coutinho 1987 [70], Li 2018 [71], Yang 1996 [72], Kettel 1996 [73] and Kettel 1998 [74] are not randomised controlled trials. We excluded Spitz 2009 because we could not confirm that it was a randomised controlled trial. The study was reported only as a book chapter; no published abstract or full text was available, and we found no information on randomisation. Mettler 1987 [75], Jiang 2001 [76], Nieto 1996 [77], Yang 2006 [78], Zhang 2016 [79], Zhao 2008 [80] and Xue 2018 [81] provided interventions that were a mixture of surgical and medical therapy. Six studies reported ineligible comparisons (Mota-González 1996 [82]; Nobel 1980 [83]; ChiCTR-TRC-10000808 [84]; Yang 2006; Xue 2016 [85]; Zhao 2013 [86]). Thomas 1987 [87] focused on an ineligible population, concentrating on asymptomatic endometriosis and its effects on fertility. Four studies were either terminated or withdrawn (NCT00556075 [88]; NCT00958412 [89]; NCT01961908 [90]; Taylor 2024 [91, 92]).

See [Supplementary material 3](#) for details about all excluded studies.

There are four studies that are awaiting classification. NCT00160433 [93] is reported as having been completed in 2008 and NCT01728454 [94] in 2017, but there are no published results. We considered two additional studies identified in the updated search to be eligible (Malavasi 2025 [95]; NCT06543550 [96, 97]), but they do not currently have sufficient outcome data available for inclusion.

One study is ongoing (EUCTR2018-002367-26-BG [98]).

See [Supplementary material 4](#) and [Supplementary material 5](#) for more information.

Risk of bias in included studies

Full details of the risk of bias assessment are presented in [Supplementary material 6](#). We assessed the risk of bias for all included studies using the RoB 2 tool and made outcome-level judgements for each prespecified outcome. The RoB 2 tool assesses risk of bias across five domains: (1) bias arising from the randomisation process; (2) bias due to deviations from intended interventions; (3) bias due to missing outcome data; (4) bias in measurement of the outcome; and (5) bias in selection of the reported result. We derived the overall risk of bias judgement for each outcome by considering bias across all five domains.

Randomisation process

Carbonell 2016 used a random list obtained from the MEDSTAT 2.1 programme and applied opaque sealed envelopes that contained a card indicating the treatment group to which the participant was assigned (low risk of bias). Carbonell 2012 used a computer-generated randomisation process and applied opaque sealed envelopes that contained a card indicating the treatment group to which the participant was assigned (low risk of bias). GISG 1996 used a double-dummy technique to conceal allocation but did not include details of the randomisation process and whether the random sequence was concealed; therefore, we had some concerns about bias. We judged Fedele 1989 and Bromham 1995 as ‘some concerns’ in domain 1, due to the absence of information on randomisation. We judged Hornstein 1990 to have ‘some concerns’ in domain 1 as authors used a block randomisation method designed to create balance across their small sample size ($n = 12$). Arguably, permuted block randomisation is problematic when applied to such a small study population.

Deviation from the intended interventions

We rated five studies as being low risk of deviation from the intended interventions (Bromham 1995; Carbonell 2012; Carbonell 2016; GISG 1996; Hornstein 1990). Fedele 1989 was judged as ‘high risk’ for domain 2, due to the absence of blinding.

Missing outcome data

Carbonell 2012, Carbonell 2016, Fedele 1989 and GISG 1996 provided outcome data for most participants and were judged to be at low risk of bias. Bromham 1995 reported large losses to follow-up (more than 10%) for all outcomes, and Hornstein 1990 did not provide data for more than 10% of participants for pain outcomes, and we judged these studies to be at high risk of bias in this domain.

GISG 1996 had a high risk of bias in domain 4 for verbal rating scores of dysmenorrhoea and dyspareunia, as the authors used the Biberoglu & Behrman questionnaire [45]. This tool has been

demonstrated to have poor validity as a pain measurement scale [99, 100].

Measurement of the outcome

We judged Carbonell 2012, Carbonell 2016, Bromham 1995, and Hornstein 1990 to be at low risk in this domain. Fedele 1989 had a high risk of bias for all outcomes in domain 4, as outcome assessors, the participants, were aware of their allocated group. GISG 1996 had a high risk of bias in domain 4 for verbal rating scores of dysmenorrhoea and dyspareunia, as the authors used the Biberoglu & Behrman questionnaire [45]. This tool has been demonstrated to have poor validity as a pain measurement scale [99, 100].

Selection of the reported result

The studies published before 2000 lack a registered trial protocol to cross-reference methodology and plans for statistical analysis and thus were judged to have ‘some concerns’ with regard to bias related to selection of the reported result (Bromham 1995; GISG 1996; Hornstein 1990; Fedele 1989). Similarly, for Carbonell 2012, there were some concerns in domain 5, due to the lack of trial protocol registration. The authors rectified this in a subsequent publication (Carbonell 2016), which is also included in this review, for which Carbonell 2012 was a pilot study. Regarding Carbonell 2016, there was a low risk of bias in all domains for outcomes related to adverse events, with the exception of raised transaminases, for which the authors did not provide a per group analysis. However, bias in domain 5 was introduced for pain outcomes (dysmenorrhoea, dyspareunia, pelvic pain) as the authors used two methods of evaluating pain (severity indicated by visual analogue scale and symptom prevalence) and appeared to focus on the favourable result.

Synthesis of results

See [Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#); [Supplementary material 7](#).

We extracted summary data from six studies (Bromham 1995; Carbonell 2012; Carbonell 2016; Fedele 1989; GISG 1996; Hornstein 1990). In studies with multiple intervention arms, we combined relevant groups for placebo-controlled meta-analyses to avoid double-counting participants in the shared comparator group, in accordance with Cochrane guidance [50]. We analysed dose comparisons as separate pairwise comparisons. Some effect estimates appeared large, which is likely due to small sample sizes and low event rates in the included studies; this is reflected in the wide confidence intervals and low certainty of the evidence. Sensitivity analyses excluding studies at high risk of bias were not possible due to the small number of studies contributing to each outcome. Results are presented below.

1. PRM versus placebo

Two included studies compared a PRM with placebo (Carbonell 2016; Chwalisz 2004). Carbonell 2016 compared three doses of mifepristone (2.5 mg, 5 mg, 10 mg) versus placebo for three months and also provided within-study dose comparisons at six months. Chwalisz 2004 compared three different doses of asoprisnil versus placebo and reported the average daily combined non-menstrual pelvic pain/dysmenorrhoea scores during all treatment months of three different doses of asoprisnil compared to placebo. However, the study did not provide extractable numerical data for the

prespecified outcomes; therefore, it was not included in the main analysis and instead is presented in [Table 1](#).

1.1 Mifepristone versus placebo

Critical outcomes

1.1.1 Overall pain

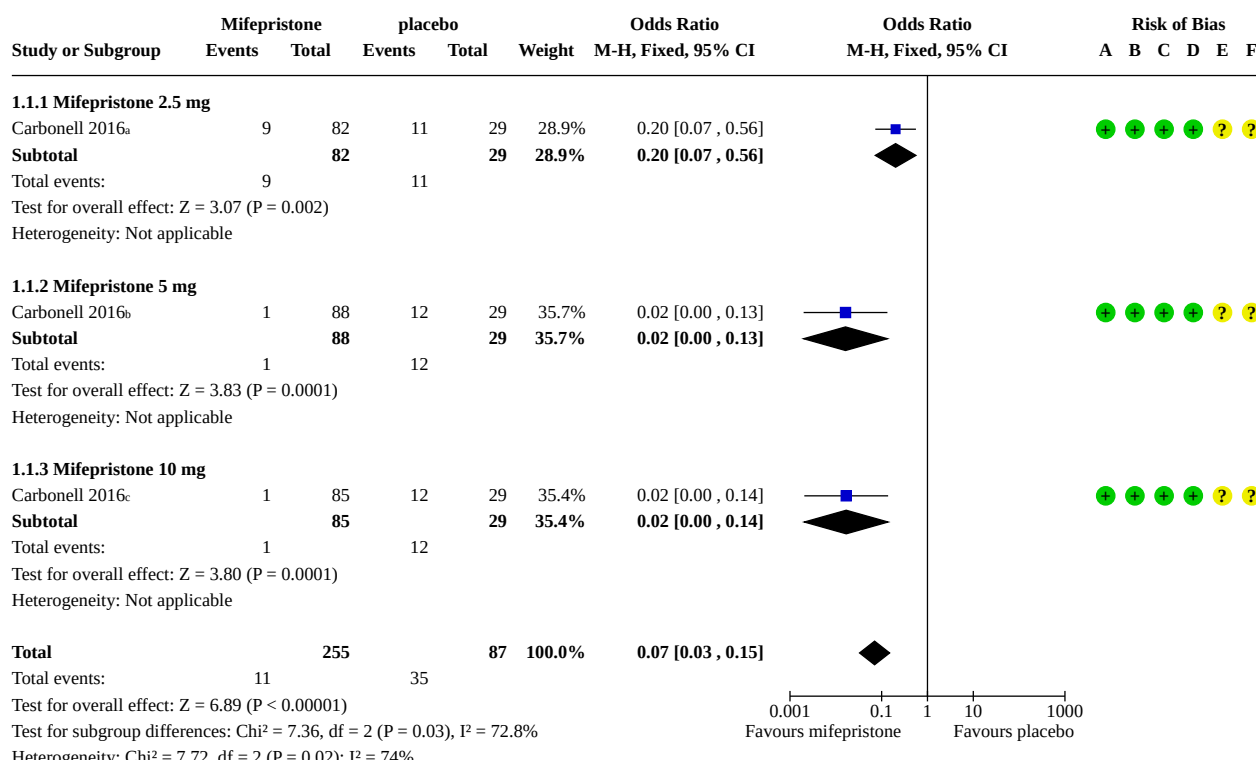
This outcome was not reported in any of the included studies.

1.1.2 Improvement in the most troublesome symptom

Dysmenorrhoea

One study reported the presence of dysmenorrhoea at three months among women who were symptomatic at baseline (Carbonell 2016). At three months, the mifepristone group may improve dysmenorrhoea compared to placebo (OR 0.07, 95% CI 0.03 to 0.15; $I^2 = 74\%$; 1 study, 342 participants; [Figure 2](#); low-certainty evidence).

Figure 2. Forest plot of comparison: 1 Effectiveness of mifepristone versus placebo, patient-assessed outcomes, outcome: 1.1 Dysmenorrhoea at three months.



Footnotes

^aIn women with dysmenorrhoea at baseline. Placebo group (total 35/97) divided. Outcome only apply to a selected subgroup of participants (114/360) women randomized.

^bIn women with dysmenorrhoea at baseline. Placebo group (total 35/97) divided. Outcome only apply to a selected subgroup of participants (120/360) women randomized.

^cIn women with dysmenorrhoea at baseline. Placebo group (total 35/97) divided. Outcome only apply to a selected subgroup of participants (118/360) women randomized.

Risk of bias legend

- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

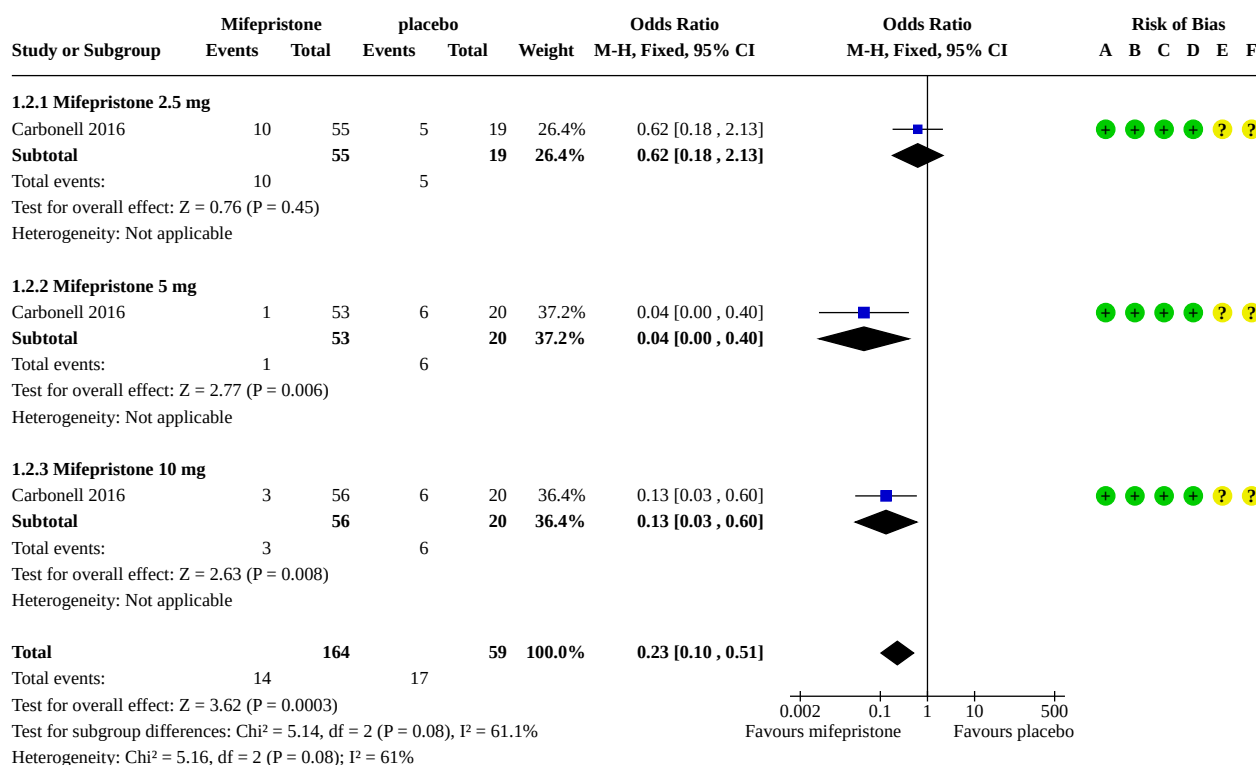
Carbonell 2016 included three mifepristone dose groups (2.5 mg, 5 mg and 10 mg). Because these dose comparisons were derived from a single study, we did not present them as formal subgroup analyses in the main text. Narratively, all three doses appear to reduce dysmenorrhoea compared with placebo, with a greater apparent effect for the 5 mg and 10 mg doses than for the 2.5 mg dose. There is no clear difference between the 5 mg and 10 mg

doses. Detailed dose data are presented in [Supplementary material 7](#).

Dyspareunia

We are uncertain whether mifepristone improves dyspareunia compared to placebo (OR 0.23, 95% CI 0.10 to 0.51; $I^2 = 61\%$; 1 study, 223 participants; [Figure 3](#); very low-certainty evidence).

Figure 3.

**Risk of bias legend**

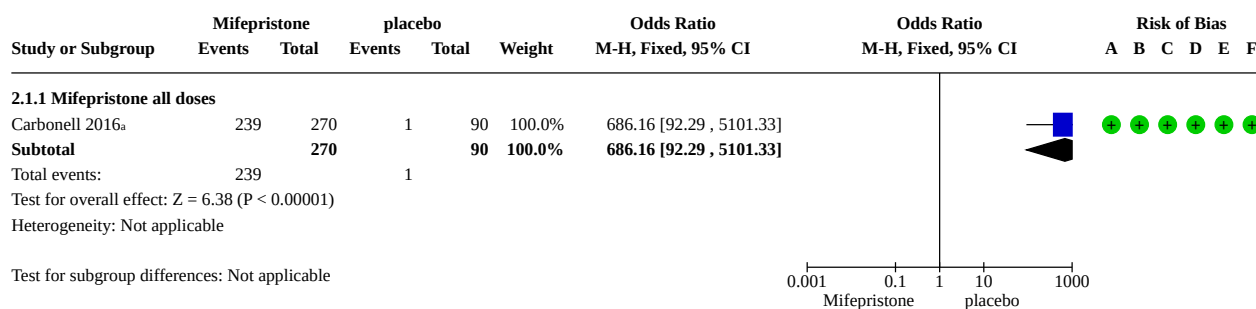
- (A) Bias arising from the randomization process
- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Carbonell 2016 included three mifepristone dose groups (2.5 mg, 5 mg and 10 mg). Because these dose comparisons were derived from a single study, we did not present them as formal subgroup analyses in the main text. Narratively, the 5 mg and 10 mg doses appear to reduce dyspareunia compared with placebo, whereas the 2.5 mg dose shows little or no effect, but the evidence is very uncertain. There was no clear difference between the 5 mg and 10 mg doses. Detailed dose data are presented in [Supplementary material 7](#).

1.1.3 Adverse events

The mifepristone group may have higher rates of certain adverse events, in terms of prevalence of amenorrhoea (OR 686.16, 95% CI

92.29 to 5101.33; 1 study, 360 participants; [Figure 4](#); low-certainty evidence), hot flushes (OR 28.79, 95% CI 3.93 to 210.73; 1 study, 360 participants; [Figure 5](#); low-certainty evidence), and PRM-associated endometrial changes (OR 6.63, 95% CI 1.26 to 34.82; I² = 0%; 1 study, 282 participants; low-certainty evidence). Mifepristone may make little or no difference after three months of treatment in prevalence of raised transaminases (OR 1.01, 95% CI 0.27 to 3.82; 1 study, 353 participants; low-certainty evidence), nausea (OR 1.72, 95% CI 0.20 to 15.03; I² = 0%; 1 study, 360 participants; [Figure 6](#); low-certainty evidence), vomiting (OR 1.02, 95% CI 0.10 to 10.01; I² = 0%; 1 study, 360 participants; low-certainty evidence), or fatigue (OR 5.48, 95% CI 0.71 to 42.27; I² = 0%; 1 study, 360 participants; low-certainty evidence). No studies reported on other adverse events.

Figure 4. Forest plot of comparison. 2 Mifepristone versus placebo, patient-assessed outcomes, outcome: 2.1 Amenorrhoea at three months.**Footnotes**^aPlacebo group not divided across subgroups as there was only one event.**Risk of bias legend**

- (A) Bias arising from the randomization process
 (B) Bias due to deviations from intended interventions
 (C) Bias due to missing outcome data
 (D) Bias in measurement of the outcome
 (E) Bias in selection of the reported result
 (F) Overall bias

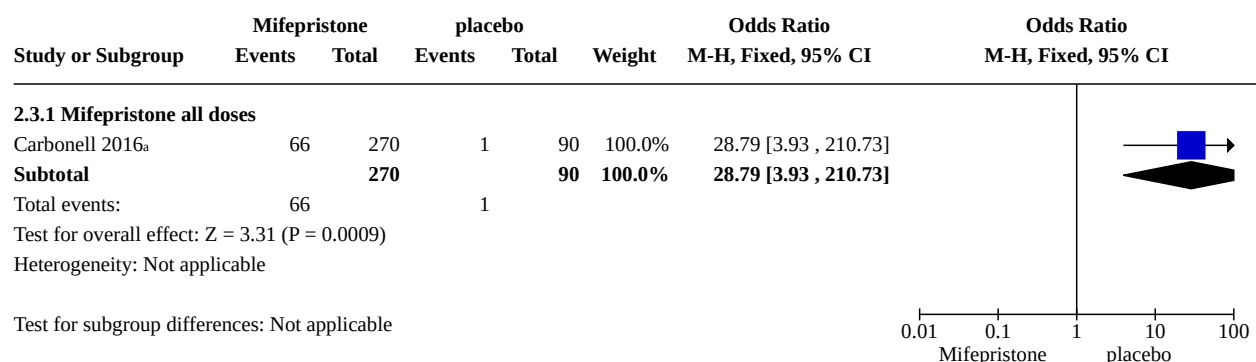
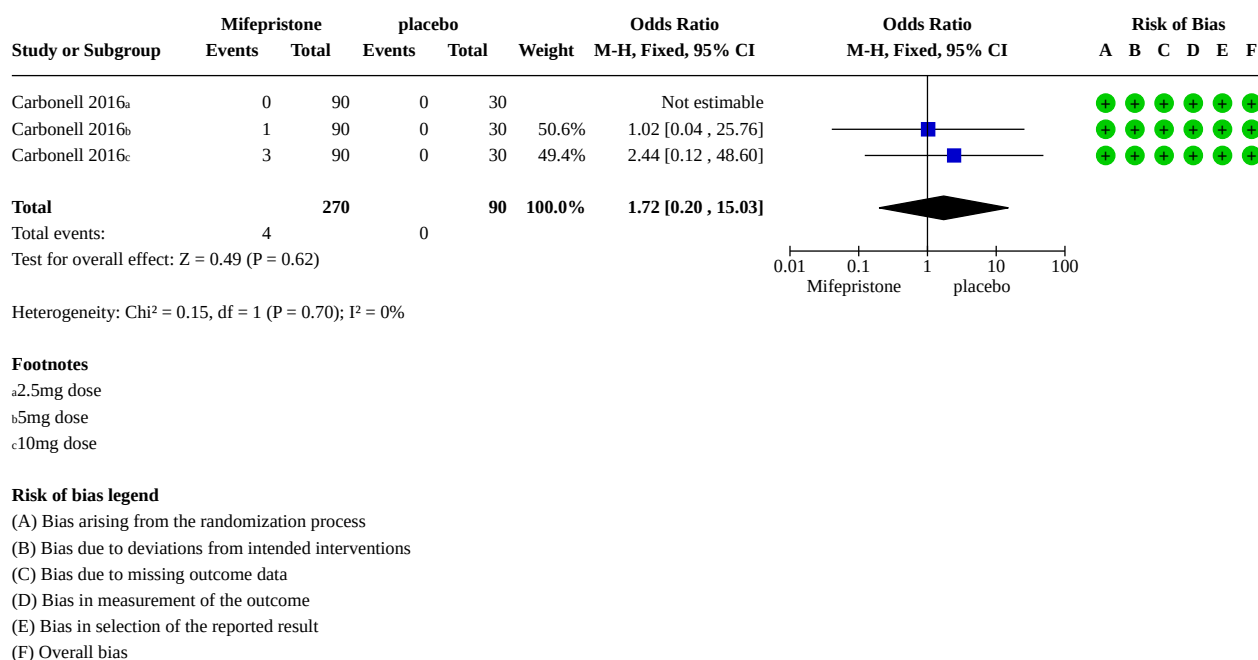
Figure 5. Forest plot of comparison: side effects at three months, mifepristone versus placebo, patient-assessed outcomes, outcome: 2.2 Hot flushes at three months.**Footnotes**^aPlacebo group not divided across subgroups as there was only one event

Figure 6.



Carbonell 2016 reported adverse events separately for the three mifepristone dose groups. Because these dose comparisons were derived from a single study, we did not present them as formal subgroup analyses in the main text. Narratively, amenorrhoea appeared more frequently reported with the 5 mg and 10 mg doses than with the 2.5 mg dose. For other adverse events, including hot flushes, raised transaminases, PRM-associated endometrial changes, nausea, vomiting, and fatigue or tiredness, there was no clear evidence of a dose-response pattern. Detailed dose data are presented in [Supplementary material 7](#).

Important outcomes

1.1.4 Quality of life

This outcome was not reported in any of the included studies.

1.1.5 Patient satisfaction with treatment

This outcome was not reported in any of the included studies.

2. PRM versus no treatment

No study reported on this comparison.

3. PRM versus other medical treatment

We identified three studies that reported on this comparison (Bromham 1995; Fedele 1989; GISG 1996). Bromham 1995 and Fedele 1989 compared gestrinone to danazol for six months of treatment. GISG 1996 compared gestrinone with the GnRH analogue leuporelin for six months.

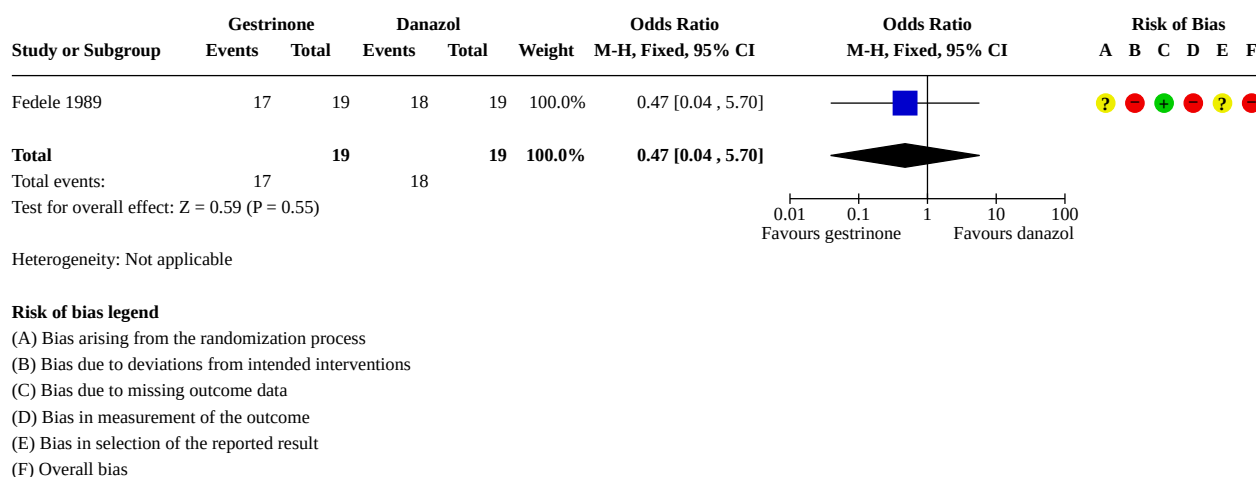
3.1 Gestrinone versus danazol

Critical outcomes

3.1.1 Overall pain

One study reported on this outcome (Fedele 1989). We are uncertain whether gestrinone provides any benefit over danazol in the improvement of overall pain (we counted as 'improved' those reporting no or mild pelvic pain) (OR 0.47, 95% CI 0.04 to 5.70; 1 study, 38 participants; [Figure 7](#); very low-certainty evidence). We judged the one study that reported on this outcome to be at high risk of bias, but given the scarcity of studies, sensitivity analysis was not possible.

Figure 7.



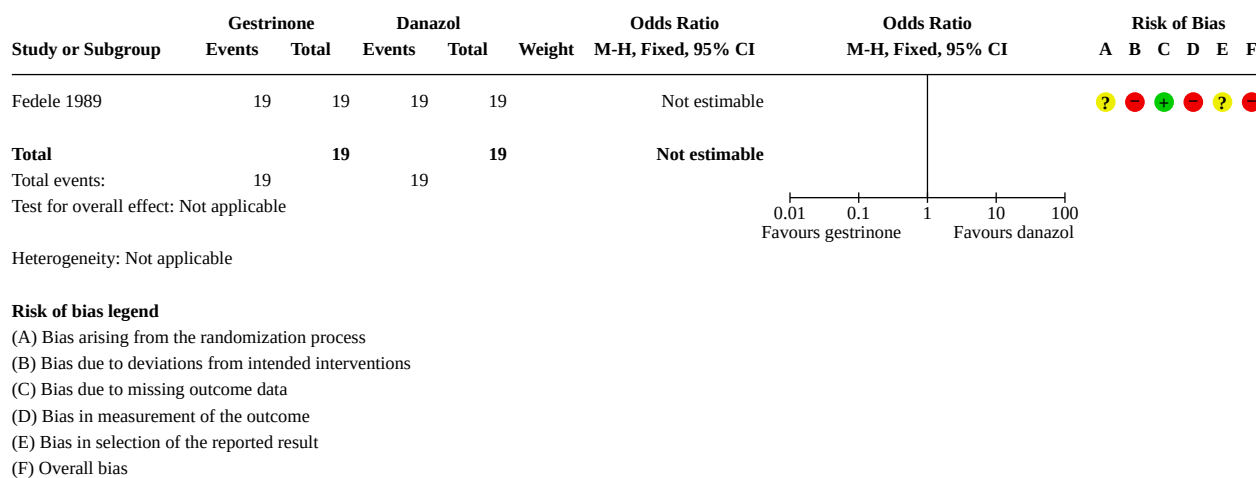
3.1.2 Improvement in the most troublesome symptom

Dysmenorrhoea

One study reported on this outcome (Fedele 1989). Effect estimates could not be calculated because all participants in both groups

reported improvement in dysmenorrhoea. We are therefore uncertain whether gestrinone provides any benefit over danazol in the improvement of dysmenorrhoea (Not estimable; 1 study, 38 participants; [Figure 8](#); very low-certainty evidence).

Figure 8.

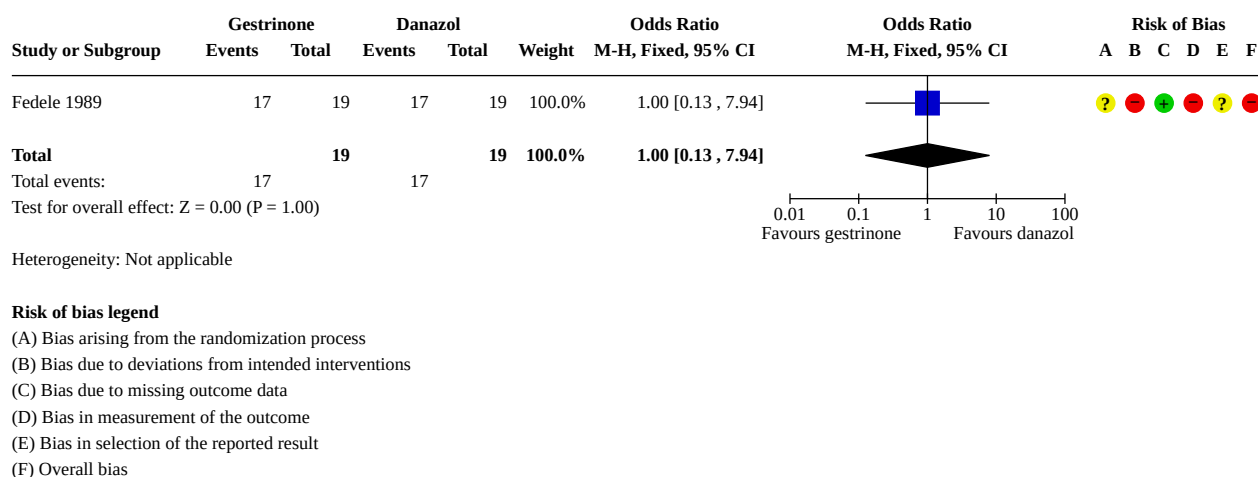


We judged the one study that reported on this outcome to be at high risk of bias, but given the scarcity of studies, sensitivity analysis was not possible.

Dyspareunia

One study reported on this outcome (Fedele 1989). We are uncertain whether gestrinone provides any benefit over danazol in the improvement of dyspareunia (OR 1.00, 95% CI 0.13 to 7.94; 1 study, 38 participants; [Figure 9](#); very low-certainty evidence).

Figure 9.



We judged the one study that reported on this outcome to be at high risk of bias, but given the scarcity of studies, sensitivity analysis was not possible.

3.1.3 Adverse events

Two studies reported on adverse events in this comparison (Bromham 1995; Fedele 1989).

We are very uncertain whether gestrinone has a lower rate of hot flushes (OR 0.79, 95% CI 0.50 to 1.26; $I^2 = 72\%$; 2 studies, 302 participants; [Figure 10](#); very low-certainty evidence), decreased breast size (OR 0.62, 95% CI 0.38 to 0.98; $I^2 = 0\%$; 2 studies, 302

participants; [Figure 10](#); very low-certainty evidence), vomiting (OR 0.67, 95% CI 0.32 to 1.43; 1 study, 264 participants; [Figure 10](#); very low-certainty evidence) or raised liver transaminases (OR 0.18, 95% CI 0.01 to 4.00; 1 study, 38 participants; [Figure 10](#); very low-certainty evidence) compared to danazol. Gestrinone may cause higher rates of hirsutism (OR 2.63, 95% CI 1.60 to 4.32; $I^2 = 68\%$; 2 studies, 302 participants; [Figure 10](#); low-certainty evidence), but there may be little or no difference in rates of acne (OR 1.45, 95% CI 0.90 to 2.33; $I^2 = 41\%$; 2 studies, 302 participants; [Figure 10](#); low-certainty evidence) or nausea (OR 1.36, 95% CI 0.84 to 2.19; $I^2 = 0\%$; 2 studies, 302 participants; [Figure 10](#); low-certainty evidence) compared to danazol.

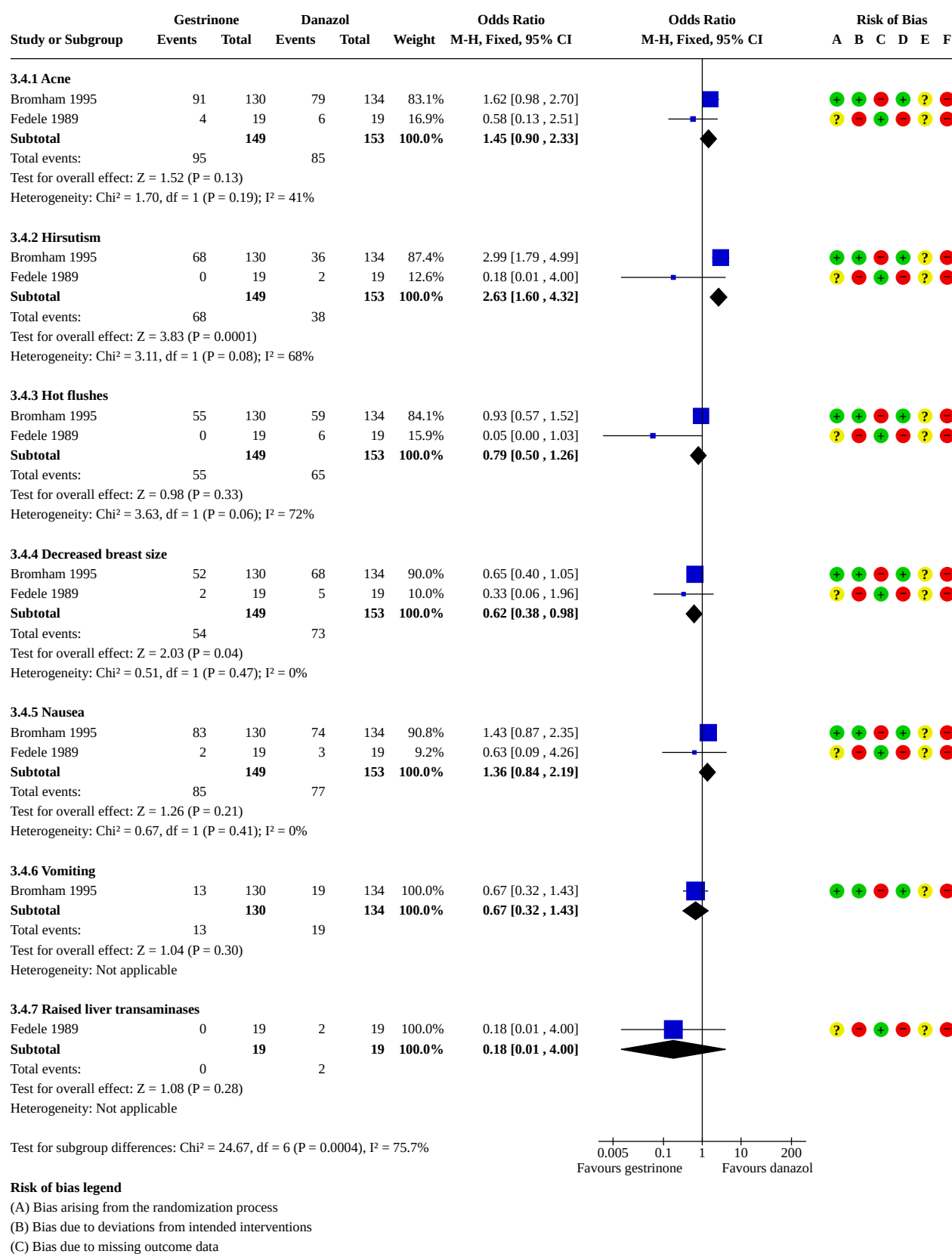
Figure 10.


Figure 10. (Continued)

- (B) Bias due to deviations from intended interventions
- (C) Bias due to missing outcome data
- (D) Bias in measurement of the outcome
- (E) Bias in selection of the reported result
- (F) Overall bias

Both studies were deemed to be at high risk of overall bias, so no sensitivity analysis was performed.

Important outcomes

3.1.4 Quality of life

This outcome was not reported in any of the included studies.

3.1.5 Patient satisfaction with treatment

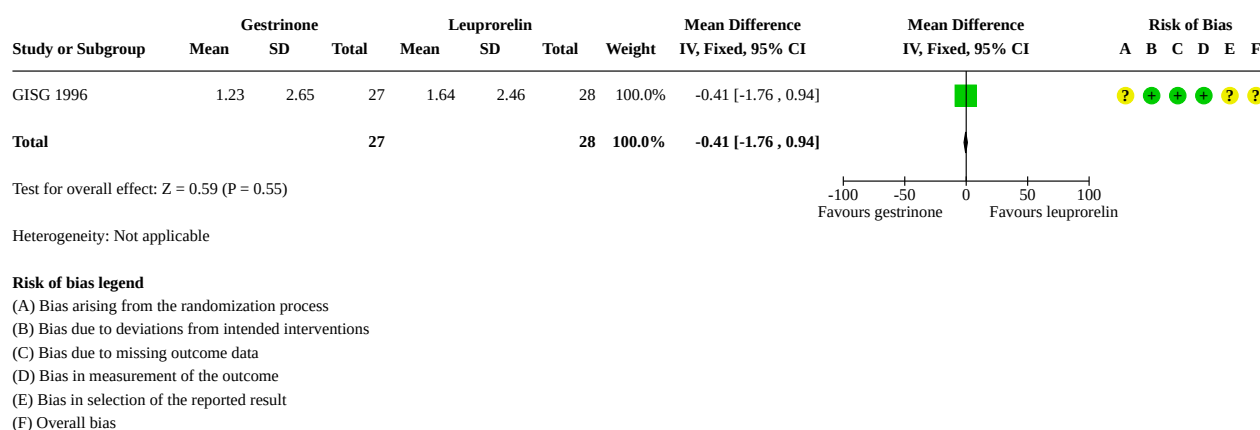
This outcome was not reported in any of the included studies.

3.2 Gestrinone versus a GnRH analogue (leuporelin)

Critical outcomes

3.2.1 Overall pain

One study reported on this outcome (GISG 1996). The gestrinone group may show little to no difference in non-menstrual pain compared to the leuporelin group (MD -0.41, 95% CI -1.76 to 0.94; 1 study, 55 participants; [Figure 11](#), low-certainty evidence).

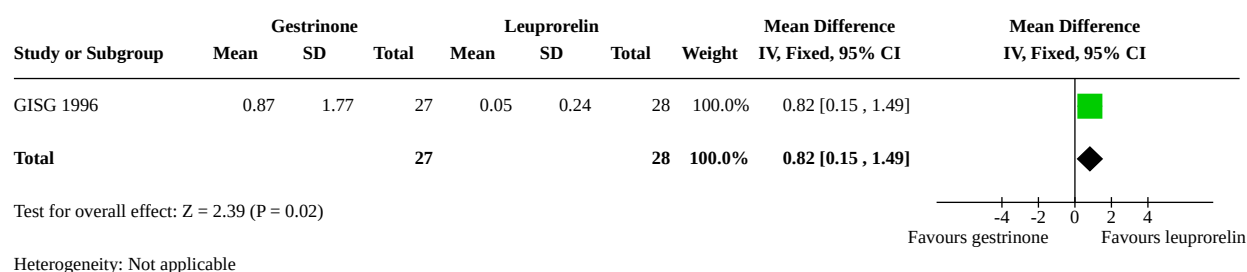
Figure 11.

We did not perform sensitivity analysis, as only one study was included for this outcome.

3.2.2 Improvement in the most troublesome symptom

Dysmenorrhoea

One study reported on this outcome (GISG 1996). The gestrinone group may have higher dysmenorrhoea scores compared to the leuporelin group (MD 0.82, 95% CI 0.15 to 1.49; 1 study, 55 participants; [Figure 12](#), low-certainty evidence).

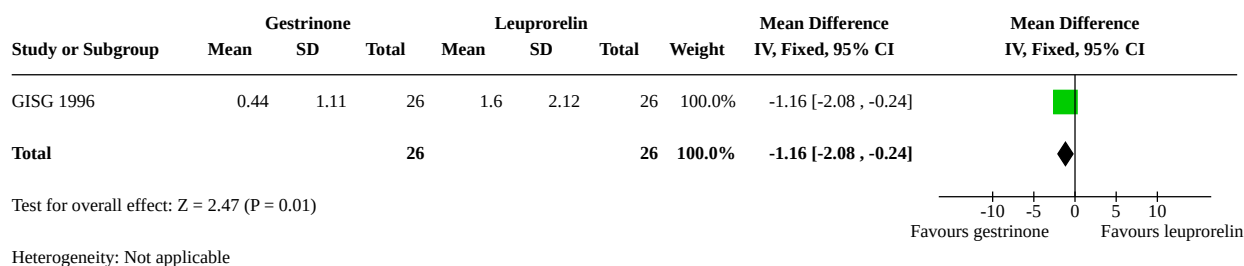
Figure 12. Forest plot of comparison: 7 Gestrinone versus leuprolin for six months: efficacy and side effects, outcome: 7.1 Painful periods, visual analogue scale.

We did not perform sensitivity analysis, as only one study was included for this outcome.

Dyspareunia

One study reported on this outcome (GISG 1996). Gestrinone may improve dyspareunia scores compared to leuporelin (MD -1.16, 95% CI -2.08 to -0.24; 1 study, 52 participants; [Figure 13](#); low-certainty evidence).

Figure 13. Forest plot of comparison: 7 Gestrinone versus leuprolin for six months: efficacy and side effects, outcome: 7.3 Pain on intercourse, visual analogue scale.



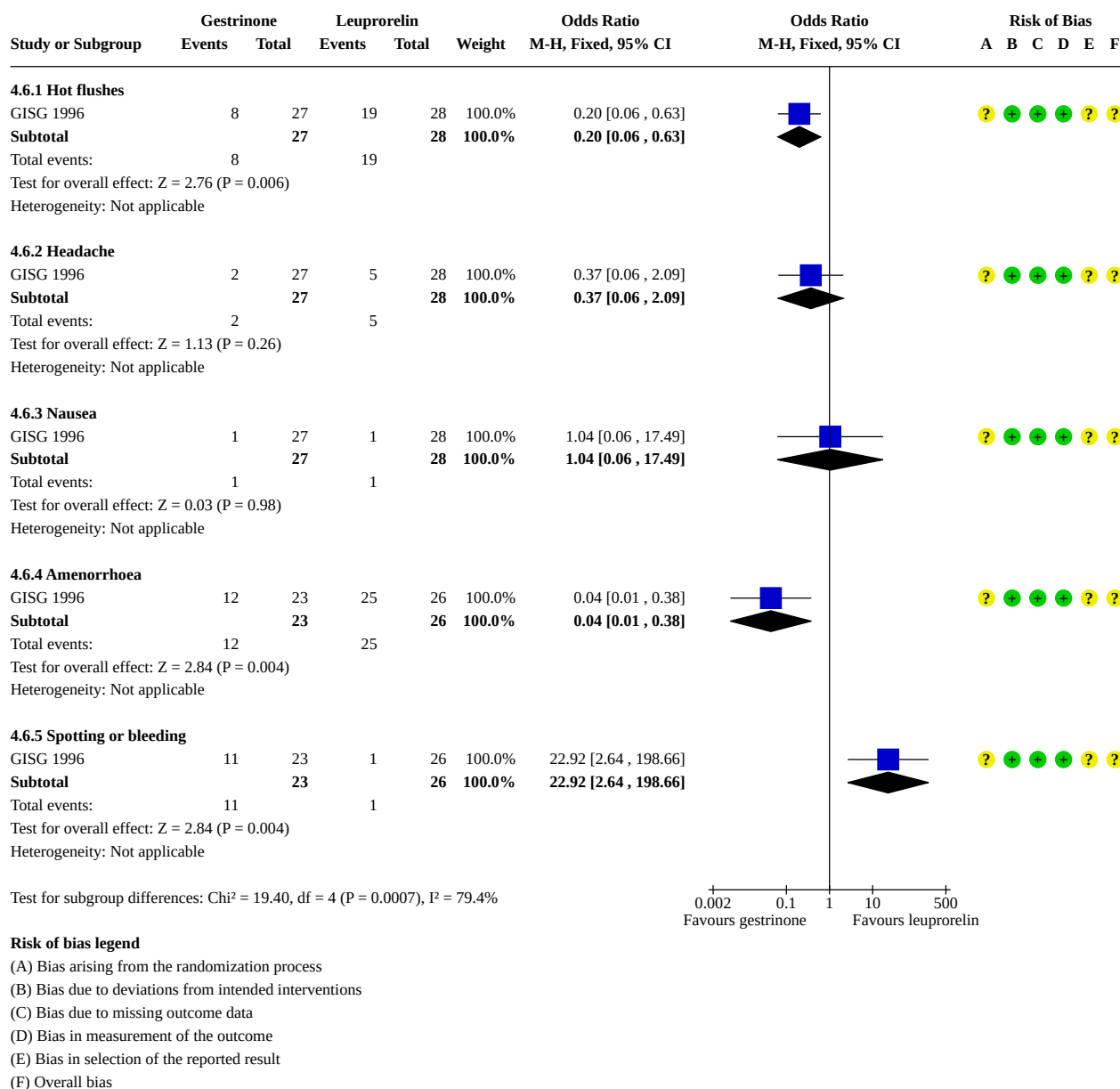
We did not perform sensitivity analysis, as only one study was included for this outcome.

3.2.3 Adverse events

One study reported on adverse events (GISG 1996). The gestrinone group may have lower rates of amenorrhoea (OR 0.04, 95% CI 0.01 to 0.38; 1 study, 49 participants; [Figure 14](#); low-certainty evidence) and hot flushes (OR 0.20, 95% CI 0.06 to 0.63; 1 study,

55 participants; [Figure 14](#); low-certainty evidence), but higher rates of irregular spotting or bleeding (OR 22.92, 95% CI 2.64 to 198.66; 1 study, 49 participants; [Figure 14](#); low-certainty evidence), compared with the leuprolerin group. There may be little or no difference between gestrinone and leuprolerin in rates of headache (OR 0.37, 95% CI 0.06 to 2.09; 1 study, 55 participants; [Figure 14](#); low-certainty evidence) or nausea (OR 1.04, 95% CI 0.06 to 17.49; 1 study, 55 participants; [Figure 14](#); low-certainty evidence).

Figure 14.



We did not perform sensitivity analysis, as only one study was included for this outcome.

Important outcomes

3.2.4 Quality of life

This outcome was not reported in any of the included studies.

3.2.5 Patient satisfaction with treatment

This outcome was not reported in any of the included studies.

4. PRM dose comparison

4.1 Mifepristone dose comparison

Two studies compared different doses of mifepristone at six months of follow-up. One trial compared 2.5 mg versus 5 mg and 10 mg

(Carbonell 2016), and the other compared 5 mg versus 25 mg (Carbonell 2012). Carbonell 2016 reported data on pain outcomes as well as adverse events. Carbonell 2012 provided a graph showing dysmenorrhoea and dyspareunia scores on a 0 to 10 visual analogue scale (VAS) without detailed data. We deemed the data unsuitable for meta-analysis; they are presented in Table 1. Carbonell 2012 also reported data on selected adverse events.

Critical outcomes

4.1.1 Overall pain

One study reported on this outcome (Carbonell 2016). We are uncertain whether the 2.5 mg dose provides any benefit over the 5 mg dose in terms of prevalence of overall pain (OR 1.81, 95% CI 0.63 to 5.17; 1 study, 110 participants, very low-certainty evidence). We are also uncertain whether the 5 mg dose provides any benefit

over the 10 mg dose in terms of prevalence of overall pain (OR 3.97, 95% CI 0.79 to 19.97; 1 study, 120 participants, very low-certainty evidence). We did not perform sensitivity analysis, as only one study was included for this outcome.

4.1.2 Improvement in the most troublesome symptom

Dysmenorrhoea

One study reported on this outcome (Carbonell 2016). We are uncertain whether the 2.5 mg dose provides any benefit over the 5 mg dose in terms of prevalence of dysmenorrhoea (OR 0.85, 95% CI 0.22 to 3.29; 1 study, 170 participants, very low-certainty evidence). We are also uncertain whether the 5 mg dose provides any benefit over the 10 mg dose in terms of prevalence of dysmenorrhoea (OR 1.22, 95% CI 0.32 to 4.71; 1 study, 173 participants, very low-certainty evidence). We did not perform sensitivity analysis, as only one study was included for this outcome.

Dyspareunia

One study reported on this outcome (Carbonell 2016). We are uncertain whether the 2.5 mg dose provides any benefit over the 5 mg dose in terms of prevalence of dyspareunia (OR 6.37, 95% CI 0.74 to 54.81; 1 study, 108 participants, very low-certainty evidence). We are also uncertain whether the 5 mg dose provides any benefit over the 10 mg dose in terms of prevalence of dyspareunia (OR 0.52, 95% CI 0.05 to 5.90; 1 study, 109 participants, very low-certainty evidence). We did not perform sensitivity analysis, as only one study was included for this outcome.

4.1.3 Adverse events

Two studies reported on adverse events (Carbonell 2012; Carbonell 2016).

The 2.5 mg dose may be associated with lower prevalence of amenorrhoea (OR 0.45, 95% CI 0.21 to 0.97; 1 study, 180 participants; low-certainty evidence), but there may be little or no difference in the prevalence of hot flushes (OR 0.84, 95% CI 0.38 to 1.89; 1 study, 180 participants; low-certainty evidence), nausea (OR 2.02, 95% CI 0.18 to 22.71; 1 study, 180 participants; low-certainty evidence), vomiting (OR 3.03, 95% CI 0.12 to 75.46; 1 study, 180 participants; low-certainty evidence), fatigue/tiredness (OR 13.92, 95% CI 0.77 to 250.94; 1 study, 180 participants, low-certainty evidence) or progesterone receptor modulator-associated endometrial changes (PAEC) (OR 2.27, 95% CI 0.88 to 5.86; 1 study, 148 participants, low-certainty evidence) over the 5 mg dose.

The 5 mg dose may be little or no different to the 10 mg dose in the prevalence of amenorrhoea (OR 1.10, 95% CI 0.47 to 2.56; 1 study, 180 participants, low-certainty evidence), hot flushes (OR 0.75, 95% CI 0.35 to 1.58; 1 study, 180 participants, low-certainty evidence), nausea (OR 1.00, 95% CI 0.06 to 16.24; 1 study, 180 participants; low-certainty evidence), vomiting (OR 0.33, 95% CI 0.01 to 8.20; 1 study, 180 participants; low-certainty evidence) or PAEC (OR 0.72, 95% CI 0.27 to 1.94; 1 study, 159 participants; low-certainty evidence), but may improve the prevalence of fatigue/tiredness (OR 0.03, 95% CI 0.00 to 0.54; 1 study, 180 participants; low-certainty evidence).

The 5 mg dose may be little or no different to the 25 mg dose in the prevalence of amenorrhoea (OR 0.61, 95% CI 0.08 to 4.41; 1 study, 26 participants, low-certainty evidence), hot flushes (OR 3.24, 95% CI 0.12 to 87.13; 1 study, 26 participants; low-certainty evidence)

or endometrial thickness > 20 mm (OR 1.41, 95% CI 0.28 to 7.13; 1 study, 26 participants; low-certainty evidence).

Both studies were deemed as low risk of bias, so no sensitivity analysis was performed.

Important outcomes

4.1.4 Quality of life

This outcome was not reported in any of the included studies.

4.1.5 Patient satisfaction with treatment

This outcome was not reported in any of the included studies.

4.2 Gestrinone dose comparison

One study compared two doses of gestrinone (1.25 mg twice-weekly and 2.5 mg twice-weekly) for a period of six months (Hornstein 1990).

Critical outcomes

4.2.1 Overall pain

We are uncertain whether the 1.25 mg dose provides any benefit over the 2.5 mg dose in the improvement of overall pain (OR 0.27, 95% CI 0.01 to 8.46; 1 study, 10 participants; very low-certainty evidence).

We did not perform sensitivity analysis, as only one study was included for this outcome.

4.2.2 Improvement in the most troublesome symptom

This outcome was not reported in any of the included studies.

4.2.3 Adverse events

There may be little or no difference between the 1.25 mg and 2.5 mg doses in the prevalence of any adverse event (OR 0.04, 95% CI 0.00 to 1.12; 1 study, 12 participants; low-certainty evidence), irregular bleeding (OR 2.50, 95% CI 0.16 to 38.60; 1 study, 12 participants; low-certainty evidence) or discontinuation due to headaches (OR 0.28, 95% CI 0.01 to 8.42; 1 study, 12 participants; low-certainty evidence).

We did not perform sensitivity analysis, as only one study was included for this outcome.

Important outcomes

4.2.4 Quality of life

This outcome was not reported in any of the included studies.

4.2.5 Patient satisfaction with treatment

This outcome was not reported in any of the included studies.

Reporting biases

We did not formally assess non-reporting bias using funnel plots or tests for funnel plot asymmetry because fewer than 10 studies contributed to each meta-analysis.

DISCUSSION

Summary of main results

Our key findings are summarised in [Summary of findings 1](#); [Summary of findings 2](#); [Summary of findings 3](#).

This Cochrane review aimed to determine the effectiveness and safety of progesterone receptor modulators (PRMs) versus placebo/ no treatment or other treatments for endometriosis, specifically medical therapy. We also tried to establish whether there is a dose effect of the interventions on the review's prespecified outcomes. Our searches revealed no studies comparing PRMs to no treatment. This review thus focusses entirely on PRM therapy versus placebo or other medical treatment.

We included nine randomised controlled trials (RCTs) with 922 participants in this systematic review. Our findings suggest that mifepristone may improve dysmenorrhoea in women with endometriosis compared to placebo, but the evidence is very uncertain regarding its effect on dyspareunia. Mifepristone might increase the prevalence of adverse events, including amenorrhoea, hot flushes, and PRM-associated endometrial changes. We are uncertain regarding the dose effect of mifepristone on pain outcomes. However, the 2.5 mg daily dose may result in a lower prevalence of amenorrhoea compared to the 5 mg daily dose, and the 5 mg dose may lead to a lower prevalence of fatigue compared to the 10 mg dose. There may be little or no difference in other adverse events.

Gestrinone may be less effective than the gonadotropin-releasing hormone (GnRH) analogue leuporelin in treating dysmenorrhoea, but it may improve dyspareunia. It may be associated with lower rates of amenorrhoea and hot flushes, but higher rates of irregular bleeding. We are uncertain about the effect of gestrinone compared to danazol on pain outcomes. Compared to danazol, gestrinone may have higher rates of hirsutism and cause little to no difference in rates of acne and nausea. We are uncertain about the effect of gestrinone on other adverse events, compared to danazol. We are also uncertain about the dose effect of gestrinone.

We found insufficient evidence to permit firm conclusions about the safety and effectiveness of other progesterone receptor modulators. There was one study comparing asoprisnil to placebo, but no extractable numerical outcome data were available.

No studies reported on the effect of PRMs on quality of life, patient satisfaction with treatment, or improvement of endometriosis-related symptoms other than pain.

Limitations of the evidence included in the review

Evidence on the effect of PRMs in the management of endometriosis continues to be limited. Compared to the previous version of this review [101], published in 2017, there are no additional RCTs for inclusion in the review, and there is only one ongoing trial, which is comparing two doses of mifepristone to placebo.

We prepared summary of findings tables using GRADEpro and Cochrane methods. We judged the certainty of evidence for the comparisons included in the main analysis as low to very low certainty.

A strength of this review is that all the included studies involved premenopausal women with symptoms of endometriosis. The diagnosis of endometriosis was made by direct visualisation (laparoscopically diagnosed endometriosis) and all participants were symptomatic at baseline.

The main limitations in the evidence are serious risk of bias (associated with poor reporting of methods and high or unclear rates of attrition in most studies), very serious imprecision with low event rates, small sample sizes, and very wide confidence intervals for most comparisons, and indirectness (outcomes assessed in a selected subgroup of participants). Some effect estimates were very large; however, these findings should be interpreted with caution as they are based on small studies with low event rates, leading to imprecision. Only two studies were at low risk of attrition bias, which is a matter of concern because attrition may well have been related to participant response. Results show a lack of consistency in outcome measures used, which leads to difficulties in combining data in a suitable meta-analysis, thus making it difficult for review authors to draw clinically relevant conclusions. Some studies use different evaluation methods to assess pain outcomes and appear to report on the favourable result, while others use methods with demonstrated poor validity.

In summary, the included studies partially answered the review question. Evidence is limited by the small number of comparisons reported and by the small sample size for each comparison. Studies investigated relevant participants and interventions, but future studies must be more unified and specific about study outcomes, as per the consensus regarding core endometriosis outcomes [44]. Methods of outcome evaluation differed amongst studies. There was no evidence regarding newly developed selective progesterone receptor modulators (SPRMs), such as vilaprisan or telapristone acetate.

Limitations of the review processes

Publication bias

We aimed to reduce the risk of publication bias by conducting systematic searches of multiple databases, as well as trial registries to identify unpublished and ongoing studies. We contacted all trial authors for additional information or clarification where necessary; however, either data were not available or we received no author response. It is possible that our searches did not identify all unpublished studies.

We were unable to conduct a funnel plot given the small number of included studies.

Outcome selection

Amenorrhoea, although presented as an adverse event in this review, might be a desirable outcome for some patients with endometriosis.

Our choice of multiple measures for primary outcomes (e.g. dysmenorrhoea, dyspareunia) could be another potential source of bias, especially as selective reporting is apparent in at least one of the included studies.

Agreements and disagreements with other studies or reviews

Our review's findings agree with other Cochrane reviews. Brown and colleagues agreed with our findings that GnRH analogues may improve dysmenorrhoea scores after six months of treatment over gestrinone, but may have worse dysmenorrhoea scores and increased rate of hot flushes [15]. Their review also agreed that there is uncertainty regarding the effect of gestrinone over danazol on pain outcomes, and that gestrinone may cause higher rates of hirsutism. Veth and colleagues included the comparison of gestrinone to GnRH analogues in their analysis and found there was uncertainty about the effect of gestrinone on pain outcomes [16]. Their findings agreed that there may be a higher rate of hot flushes with GnRH analogues, and they also noted that there may be decreased bone density with the use of monthly leuporelin compared to gestrinone after six months, which was not included in our prespecified outcomes. Murji and colleagues also agreed that PRM-associated endometrial changes (PAEC) in women treated with SPRMs are more frequent than in those treated with placebo [32].

The conclusions of other non-Cochrane reviews [22, 29, 102] and of studies that were excluded from this review were substantially the same as the conclusions of this review, and more studies are needed to evaluate the effectiveness and safety of PRMs. In their review, Guo and colleagues also provide valuable insights as to why studies on the effect of more newly developed SPRMs, such as asoprisnil and telapristone acetate, were either terminated or withdrawn [22].

Lastly, our review agrees with other systematic reviews, suggesting that gestrinone is associated with certain adverse events similar to danazol, including acne and hirsutism, possibly related to the reduction in serum hormone binding levels, as well as hot flushes and amenorrhoea, related to hypo-oestrogenism [103].

AUTHORS' CONCLUSIONS

Implications for practice

Given the small number of studies, limited sample sizes, and low or very low certainty of the evidence for many outcomes, we are uncertain about the overall value of progesterone receptor modulators (PRMs) in treating endometriosis-associated symptoms. Amongst women with endometriosis, low-certainty evidence suggests that mifepristone may improve dysmenorrhoea; however, the evidence is very uncertain regarding its effect on dyspareunia. Certain adverse events, such as hot flushes, may be more common with mifepristone (low-certainty evidence). Data on dosage were inconclusive. Gestrinone may be less effective than gonadotropin-releasing hormone (GnRH) analogues for dysmenorrhoea, but may improve dyspareunia (low-certainty evidence). It may cause fewer hot flushes and may result in little or no difference in nausea (low-certainty evidence). The effects of gestrinone compared with danazol or at different doses are very uncertain (very low-certainty evidence). Evidence for other progesterone receptor modulators is insufficient to draw reliable conclusions.

Implications for research

Effects of PRM-associated endometrial change on endometriotic lesions are unclear, and long-term effects remain to be determined.

This systematic review has identified the need for larger, well-designed randomised controlled trials with minimal attrition to confirm and extend the findings of the trials reviewed here. Studies comparing the relative efficacy and safety of different PRMs, including newly developed ones, against placebo and currently established medical treatment would provide important clinical evidence. Future studies should report quality of life and use validated pain relief scores. Given the concerns regarding liver toxicity with the use of ulipristal acetate and telapristone acetate, future studies studying any selective progesterone receptor modulators (SPRMs) should also focus on this adverse event. Longer use (more than six months) and more objective measurements, such as endometrial biopsy, can assess PRM-associated endometrial changes more accurately and can shed more light on the long-term safety of SPRMs. Future studies should also evaluate fertility and pregnancy outcomes in women with endometriosis who wish to conceive following treatment with progesterone receptor modulators. Given the ovulation-suppressing and antiproliferative effects of these agents, their role in the management of endometriosis-associated infertility remains uncertain. Lastly, given the evidence that fibrotic lesions associated with deep endometriosis have reduced progesterone receptor (PR) expression, hence possibly more resistant to SPRMs, future studies should stratify patients and report pain outcomes based on endometriosis severity.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD009881.pub3](https://doi.org/10.1002/14651858.CD009881.pub3).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Risk of bias

Supplementary material 7 Analyses

Supplementary material 8 Data package

Supplementary material 9 Changes between protocol and review

Supplementary material 10 Summary of findings table-Mifepristone dose comparison for endometriosis

Supplementary material 11 Summary of findings table-Gestrinone dose comparison for endometriosis

ADDITIONAL INFORMATION

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Editorial and peer-reviewer contributions

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The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Madelon van Wely, AmsterdamUMC
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Anne-Marie Stephani, Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, supported editorial team): Joshua Guinoo, Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Kerry Dwan, University of York (methods review); Jo Platt, Central Editorial Information Specialist (search review)

Contributions of authors

Selection of studies: Alexandros Loukas Grammatidis, Tatjana Gibbons

Data extraction and management: Alexandros Loukas Grammatidis, Tatjana Gibbons

Assessment of risk of bias in included studies: Alexandros Loukas Grammatidis, Eberechi N Anucha

Consultation: Christian M Becker

Review writing: Alexandros Loukas Grammatidis, Tatjana Gibbons, Eberechi N Anucha, Christian M Becker

Declarations of interest

ALG has no interests to declare.

TG has no interests to declare.

What's new

Date	Event	Description
16 September 2026	New citation required but conclusions have not changed	New citation required following updated searches, revised analyses, updated risk of bias assessment, and methodological updates. Overall conclusions remain unchanged.
16 September 2026	New search has been performed	Updated searches conducted to 16 April 2026. Nine included studies (922 participants); one previously included study (Bromham 1995) was excluded from quantitative synthesis be-

ENA has no interests to declare.

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External sources

- No sources of support provided

Registration and protocol

Protocol (2012): <https://doi.org/10.1002/14651858.CD009881>

Review (2017): <https://doi.org/10.1002/14651858.CD009881.pub2>

Data, code and other materials

As part of the published Cochrane review, the following is made available for download for users of the Cochrane Library (see [Supplementary material 8](#)): full search strategies for each database; full citations of each unique report for all studies (included, ongoing or awaiting classification, or excluded at the full text screen) in the final review; study data, including study information, study results or test data; consensus risk of bias assessments; and analysis data [104]. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods [105]. Template forms assessing RoB from Excel are available from the authors on reasonable request.

Additional materials accompanying this review are available as [Supplementary material 1](#) (search strategies), [Supplementary material 2](#) (characteristics of included studies), [Supplementary material 3](#) (characteristics of excluded studies), [Supplementary material 4](#) (characteristics of studies awaiting classification), [Supplementary material 5](#) (characteristics of ongoing studies), [Supplementary material 6](#) (risk of bias), [Supplementary material 7](#) (analyses), [Supplementary material 8](#) (data package), [Supplementary material 9](#) (changes between protocol and review), [Supplementary material 10](#) (summary of findings table – mifepristone dose comparison for endometriosis), and [Supplementary material 11](#) (summary of findings table – gestrinone dose comparison for endometriosis).

Date	Event	Description
		cause outcome data were not extractable. Review methods were updated in line with current Cochrane guidance.

History

Protocol first published: Issue 5, 2012

Review first published: Issue 7, 2017

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ADDITIONAL TABLES

Table 1. Data unsuitable for meta-analysis

Outcome	Study	Comparison	Measure	Intervention group	Control group	P value
Combined non-pelvic pain, dysmenorrhoea, and dyspareunia	Chwalisz 2004	Asoprisnil: 5 mg (n = 31), 10 mg (n = 33), 25 mg (n = 32) Placebo (n = 34)	Mean reduction at 3 months on 0 to 4 pain scale	0.5 points for each dose	< 0.1 points	< 0.05
Lower abdominal pain	Bromham 1995	Gestrinone: 2.5 mg twice weekly (n = 132), Danazol 200 mg twice daily (n = 137)	Reduction in severity at 3 months	Decreased severity (no numerical value given)	Decrease in severity (no numerical value given)	< 0.0001
Dysmenorrhoea	Bromham 1995	Gestrinone: 2.5 mg twice weekly (n = 132), Danazol 200 mg twice daily (n = 137)	Reduction in severity at 6 months	Decreased severity (no numerical value given)	Decreased severity (no numerical value given)	< 0.001
Dysmenorrhoea	Bromham 1995	Gestrinone: 2.5 mg twice weekly (n = 132), Danazol 200 mg twice daily (n = 137)	Absence of dysmenorrhoea at 6 months	Absent in 60%	Absent in 52%	N/A
Dyspareunia	Bromham 1995	Gestrinone: 2.5 mg twice weekly (n = 132), Danazol 200 mg twice daily (n = 137)	Symptom prevalence at baseline vs at 6 months	50% vs 15%	56% vs 17%	N/A
Dysmenorrhoea	Carbonell 2012	Mifepristone: 5 mg daily (n = 13), Mifepristone: 25 mg daily (n = 13)	Mean reduction in severity 0 to 10 VAS scale at 6 months	MD 0.5 lower (2.04 lower to 1.04 higher)		> 0.05
Dyspareunia	Carbonell 2012	Mifepristone: 5 mg daily (n = 13), Mifepristone: 25 mg daily (n = 13)	Mean reduction in severity 0 to 10 VAS scale at 6 months	MD 0 (1.23 lower to 1.23 higher)		> 0.05

n: number; N/A: not applicable; P: probability; VAS: visual analogue scale; vs: versus; >: less than

Table 2. Overview of included studies

Study ID	Experimental	Comparator	Outcome	Parti- pants (n)	RoB - D1	RoB - D2	RoB - D3	RoB - D4	RoB - D5	Overall RoB
Bromham 1995 ^a	Gestrinone 2.5 mg twice week- ly plus 'dummy' danazol	Danazol 200 mg twice daily plus 'dummy' gestri- none	Prevalence of side effects - hirsutism, acne, seborrhoea, hot flushes, de- creased breast size, headache, nausea, vomiting, tiredness	269	Some con- cerns	Low risk	Low risk	Low risk	Some con- cerns	Some con- cerns
Carbonell 2012	Mifepristone 25mg	Mifepristone 5mg	Prevalence of side ef- fects - amenorrhoea, hot flushes		Low risk	Low risk	Low risk	Low risk	Some con- cerns	Some con- cerns
	Mifepristone 25mg	Mifepristone 5mg	Endometrial thick- ness less than 20 mm		Low risk	Low risk	Low risk	Low risk	Some con- cerns	Some con- cerns
Carbonell 2016	Mifepristone 2.5, 5, 10 mg	Placebo	Prevalence of dys- menorrhoea	360	Low risk	Low risk	Low risk	Low risk	Some con- cerns	Some con- cerns
	Mifepristone 2.5, 5, 10 mg	Placebo	Prevalence of dys- pareunia		Low risk	Low risk	Low risk	Low risk	Some con- cerns	Some con- cerns
	Mifepristone 2.5, 5, 10 mg	Placebo	Prevalence of pelvic pain		Low risk	Low risk	Low risk	Low risk	Some con- cerns	Some con- cerns
	Mifepristone 2.5, 5, 10 mg	Placebo	Prevalence of side effects -amenor- rhoea, hot flushes, nausea, vomiting, fa- tigue/tiredness		Low risk	Low risk	Low risk	Low risk	Low risk	Low risk
Fedele 1989	Gestrinone 2.5 mg twice week- ly (increasing to 3 times a week if no amen- orrhoea by 1 month)	Danazol 600 mg per day (in- creasing to 800 mg per day if no amenorrhoea by 1 month)	Severity of dysmen- orrhoea	39	Some con- cerns	Some con- cerns	Low risk	High risk	Some con- cerns	High risk
	Gestrinone 2.5 mg twice week- ly (increasing to	Danazol 600 mg per day (in- creasing to 800	Severity of dyspareu- nia		Some con- cerns	Some con- cerns	Low risk	High risk	Some con- cerns	High risk

Table 2. Overview of included studies (Continued)

	3 times a week if no amenorrhoea by 1 month)	mg per day if no amenorrhoea by 1 month)								
	Gestrinone 2.5 mg twice weekly (increasing to 3 times a week if no amenorrhoea by 1 month)	Danazol 600 mg per day (increasing to 800 mg per day if no amenorrhoea by 1 month)	Severity of pelvic pain		Some concerns	Some concerns	Low risk	High risk	Some concerns	High risk
	Gestrinone 2.5 mg twice weekly (increasing to 3 times a week if no amenorrhoea by 1 month)	Danazol 600 mg per day (increasing to 800 mg per day if no amenorrhoea by 1 month)	Prevalence of side effects - acne, hirsutism, hot flushes, decreased breast size, nausea, raised liver transaminases		Some concerns	Some concerns	Low risk	High risk	Some concerns	High risk
GISG 1996	Gestrinone 2.5 mg twice weekly plus placebo injections	Intramuscular (IM) leuprolide acetate 3.75 mg once a month plus placebo tablets	Dysmenorrhoea - reduction in mean severity score on a visual analogue scale	55	Some concerns	Low risk	Low risk	Low risk	Some concerns	Some concerns
	Gestrinone 2.5 mg twice weekly plus placebo injections	Intramuscular (IM) leuprolide acetate 3.75 mg once a month plus placebo tablets	Dysmenorrhoea - reduction in mean severity score on a verbal rating scale (Biberoglu & Behrman [45])		Some concerns	Low risk	Low risk	High risk	Some concerns	High risk
	Gestrinone 2.5 mg twice weekly plus placebo injections	Intramuscular (IM) leuprolide acetate 3.75 mg once a month plus placebo tablets	Dyspareunia - reduction in mean severity score on a visual analogue scale		Some concerns	Low risk	Low risk	Low risk	Some concerns	Some concerns
	Gestrinone 2.5 mg twice weekly	Intramuscular (IM) leuprolide acetate 3.75 mg	Dyspareunia - reduction in mean severity score on a verbal rat-		Some concerns	Low risk	Low risk	High risk	Some concerns	High risk

Table 2. Overview of included studies (Continued)

	ly plus placebo injections	once a month plus placebo tablets	ing scale (Biberoglu & Behrman [45])						
	Gestrinone 2.5 mg twice weekly plus placebo injections	Intramuscular (IM) leuprolide acetate 3.75 mg once a month plus placebo tablets	Non-menstrual pelvic pain - reduction in mean severity score on a visual analogue scale		Some concerns	Low risk	Low risk	Low risk	Some concerns
	Gestrinone 2.5 mg twice weekly plus placebo injections	Intramuscular (IM) leuprolide acetate 3.75 mg once a month plus placebo tablets	Prevalence of side effects - hot flushes, headache, nausea, drowsiness, amenorrhoea, spotting or bleeding		Some concerns	Low risk	Low risk	Low risk	Some concerns
Hornstein 1990	Gestrinone 1.25mg twice weekly	Gestrinone 2.5mg twice weekly	Improvement in pain	12	Some concerns	Low risk	Some concerns	High risk	Some concerns
	Gestrinone 1.25mg twice weekly	Gestrinone 2.5mg twice weekly	Prevalence of side effects - 'any side effect', 'discontinuation due to headache', irregular bleeding		Some concerns	Low risk	Low risk	Low risk	Some concerns
Chwalisz 2004	No extractable data for prespecified review outcomes ^b								
Dawood 1997	No prespecified review outcomes reported ^c								
Worthington 1993	No prespecified review outcomes reported ^c								

RoB: risk of bias; D1: bias arising from the randomisation process; D2: bias due to deviations from intended interventions; D3: bias due to missing outcome data; D4: bias in measurement of the outcome; D5: bias in selection of the reported result. We assessed risk of bias using the Cochrane RoB 2 tool.

^aFor Bromham 1995, the denominators for the reported pain percentages were unclear because participant numbers varied across follow-up assessments. We therefore excluded the pain data from meta-analysis; the risk-of-bias assessment shown relates to adverse events only.

^bChwalisz 2004 reported prespecified outcomes but did not provide extractable numerical data; therefore, we did not undertake an outcome-level RoB 2 assessment.

^cDawood 1997 and Worthington 1993 did not report prespecified review outcomes; therefore, we did not undertake outcome-level RoB 2 assessments.

